MagDiff™ *MA900*

Fully Differential, Magnetic Digital Angle Sensor with Stray-Field Immunity

PRELIMINARY SPECIFICATION SUBJECT TO CHANGE

DESCRIPTION

The MA900 is a contactless, precision magnetic absolute angle position sensor. The angle is extracted from the difference of magnetic field at several sensor locations on the IC. This differential method eliminates the contribution of parasitic magnetic fields and is suitable for sensors positioned at the end-of-shaft with a simple dipole target magnet. Fast data acquisition and processing provide accurate angle measurement at speeds from 0 to more than 100’000 rpm.

The MA900 features magnetic field strength detection with programmable thresholds to allow sensing of the magnet position relative to the sensor for creation of functions such as sensing of axial movements or for diagnostics.

On-chip non-volatile memory provides storage for configuration parameters, including the reference zero angle position, ABZ encoder settings, configuration of UVW pole pair emulation and magnetic field detection thresholds.

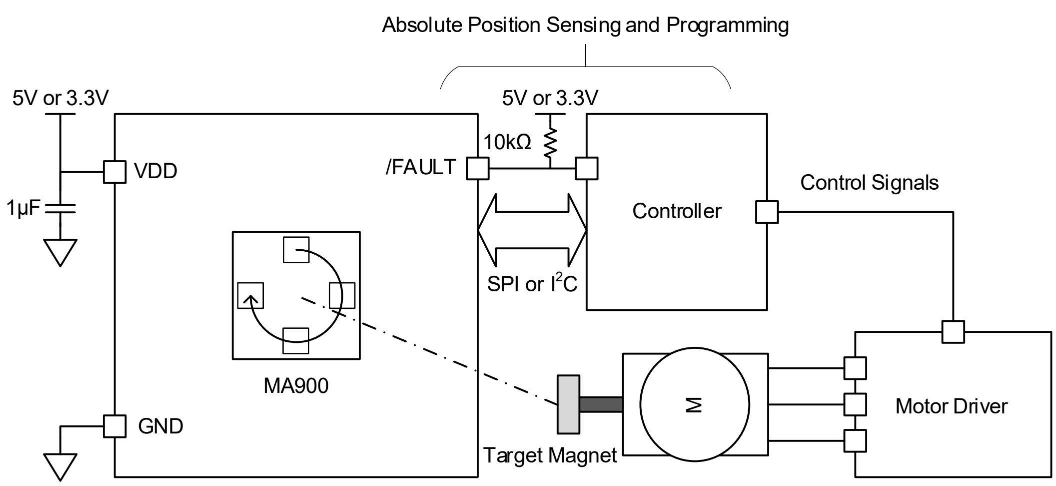
FEATURES

- Absolute Angle Encoder
- ±0.02° RMS Noise at 3kHz Bandwidth
- Robust Against Parasitic Stray Fields Exceeding 4kA/m DC, or 5mT.
- Zero Lag
- SPI or I2C Serial Interface for Digital Angle Readout and Chip Configuration
- ABZ Quadrature Encoder Interface with Programmable Pulses-Per-Turn (PPT)
- UVW Interface with 1-to-8 Pole Pair Emulation
- Programmable Magnetic Field Strength Detection for Diagnostic Checks
- Dedicated Push-Button Interface
- 3.3V & 5V Supply, 12 mA Operating Current
- -40°C to +150°C Operating Temperature
- Available in a QFN-16 (3mmx3mm) Package with Wettable Flanks

APPLICATIONS

- BLDC Absolute Motor Speed & Position Control
- Robotics
- Industrial Control & Automation

TYPICAL APPLICATION



ORDERING INFORMATION

Part Number*	Package	Top Marking	MSL Rating	Version
MA900GQE-C-0000	QFN-16 (3mm x 3mm)	*See Below*	1	I^2C - Default
MA900GQE-S-0000	QFN-16 (3mm x 3mm)	*See Below*	1	SPI - Default
MA900GQE-C-xxxx**	QFN-16 (3mm x 3mm)	*See Below*	1	I^2C - Customized Registers (contact factory)
MA900GQE-S-xxxx**	QFN-16 (3mm x 3mm)	*See Below*	1	SPI - Customized Registers (contact factory)

* For Tape & Reel, add suffix -Z (e.g., MA900GQE-C-xxxx-Z).

** "-xxxx" is the configuration code identifier for the register settings. The first four digits of the suffix ("-xxxx") can be a hexadecimal value between 0 and F. Work with an MPS FAE to create this unique number for any non-default function option. "-0000" is the factory default code.

PART NUMBERING

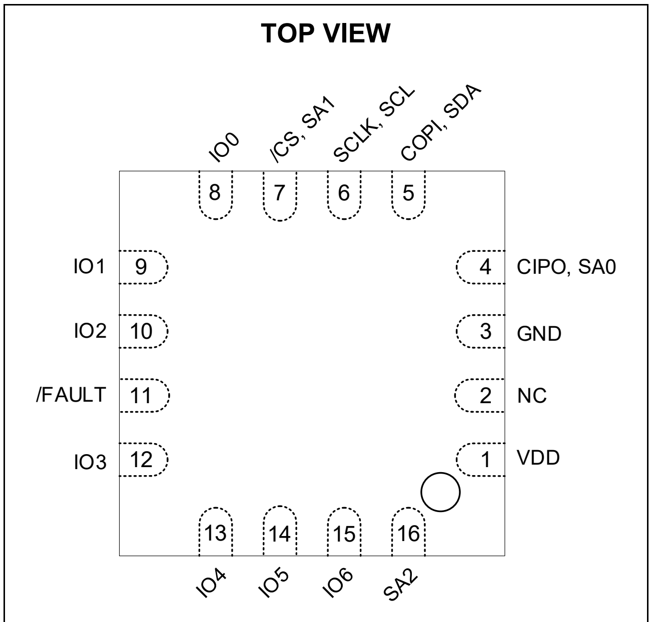
MA900GQE-y-xxxx-Z

TOP MARKING

CARY
LLLL

CAR: Product code
Y: Year code
LLLL: Lot number

PACKAGE REFERENCE



QFN-16 (3mm x 3mm)

PIN FUNCTIONS

Package Pin #	Name	Description
1	VDD	**Main Supply**. Connect to 5V or 3.3V.
2	NC	**No Connection.**
3	GND	**Supply Ground.**
4	CIPO, SA0	**SPI: Data Output**. Push-pull. High-Z when /CS is idle by default. **I²C: Serial Address Definition.** Connect to either VDD or GND.
5	COPI, SDA	**SPI: Data Input**. High impedance input. Connect to GND if not used. **I²C: Serial Data.** Open drain.
6	SCLK, SCL	**SPI: Serial Clock**. High impedance input. Connect to GND if not used. **I²C: Serial Clock**. High impedance input. Connect to GND if not used.
7	/CS, SA1	**SPI: Chip Select (Active Low).** High impedance input. Connect to VDD if not used. **I²C: Serial Address Definition.** Connect to either VDD or GND.
8	IO0	**Generic IO 0.** Functionality according to IO-Matrix. Push-pull by default. Leave floating if not used.
9	IO1	**Generic IO 1.** Functionality according to IO-Matrix. Push-pull by default. Leave floating if not used.
10	IO2	**Generic IO 2.** Functionality according to IO-Matrix. Push-pull by default. Leave floating if not used.
11	/FAULT	**Fault Output (Active Low).** Open drain. Pull-up resistor to VDD is required.
12	IO3	**Generic IO 3.** Functionality according to IO-Matrix. Push-pull by default. Leave floating if not used.
13	IO4	**Generic IO 4.** Functionality according to IO-Matrix. Push-pull by default. Leave floating if not used.
14	IO5	**Generic IO 5.** Functionality according to IO-Matrix. Push-pull by default. Leave floating if not used.
15	IO6	**Generic IO 6.** Functionality according to IO-Matrix. Push-pull by default. Leave floating if not used.
16	SA2	**SPI:** Connect to GND. **I²C: Serial Address Definition.** Connect to either VDD or GND.

ABSOLUTE MAXIMUM RATINGS (1)

Supply voltage .... -0.5V to +7.0V
Input pin voltage (V_I) .... -0.5V to +7.0V
Output pin voltage (V_O) .... -0.5V to +7.0V
Continuous power dissipation (T_A = +25°C) (2)
....2.0W
Junction temperature .... 165°C
Lead temperature .... 260°C
Storage temperature.................. -65°C to 165°C

Recommended Operating Conditions

V_{DD}, 3.3V operation.............................3V to 3.6V
V_{DD}, 5V operation.............................4.5V to 5.5V
Operating junction temp (T_J).....-40°C to +150°C

Thermal Resistance (4) *θ*

Thermal Resistance (4)	θ_{JA}	θ_{JC}	
QFN-16 (3mmx3mm)	50	12 ..	°C/W

Notes:

1) Exceeding these ratings may damage the device.
2) The maximum allowable power dissipation is a function of the maximum junction temperature T_J (MAX), the junction-to-ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by P_D (MAX) = (T_J (MAX) - T_A) / θ_{JA}. Exceeding the maximum allowable power dissipation produces an excessive die temperature, causing the regulator to go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.
3) The device is not guaranteed to function outside of its operating conditions.
4) Measured on JESD51-7, 4layer PCB.

ELECTRICAL CHARACTERISTICS

V_{DD} = 3.3V or 5V, T_A = -40°C to +150°C, typical values are at 25°C, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Recommended Operating Conditions						
Supply voltage	V_{DD}	5V operation	4.5	5	5.5	V
		3.3V operation	3	3.3	3.6	V
Supply current	I_{DD}	T_A = 25°C		12		mA
		T_A = -40°C to +125°C			18	
		T_A = -40°C to +150°C			20	
Operating Temperature	T_{op}		-40		150	°C
Operating magnetic field	B_z		8	30	200	mT
External homogeneous field	B_t				1000	mT
Digital I/O (3.3V operation)						
Input high voltage	V_{IH}		V_{DD} × 0.65		V_{DD} + 0.3	V
Input low voltage	V_{IL}		-0.3		V_{DD} × 0.35	V
Drive Current	I_O	DRV = 0		4		mA
Push-pull option						
Output low voltage	V_{OL}	DRV = 0		0.31	0.64	V
Output high voltage	V_{OH}	DRV = 0	2.60	2.90		V
Input leakage current	I_L				10	μA
Pull-up resistor	R_{PU}			63.6		kΩ
Pull-down resistor	R_{PD}			62.6		kΩ
Rising edge slew rate	T_R	DRV = 0, C_L = 50pF		0.1		V/ns
Falling edge slew rate	T_F	DRV = 0, C_L = 50pF		0.15		V/ns
Open-drain option						
Output low voltage	V_{OL}	DRV = 0			0.4	V
Open drain leakage current					10	μA
Digital I/O (5V operation)						
Input high voltage	V_{IH}		V_{DD} × 0.65		V_{DD} + 0.3	V
Input low voltage	V_{IL}		-0.3		V_{DD} × 0.35	V
Drive Current	I_O	DRV = 0		4		mA
Push-pull option						
Output low voltage	V_{OL}	DRV = 0		0.22	0.41	V
Output high voltage	V_{OH}	DRV = 0	4.57	4.70		V
Input leakage current	I_L				10	μA
Pull-up resistor	R_{PU}			39		kΩ
Pull-down resistor	R_{PD}			39		kΩ
Rising edge slew rate	T_R	DRV = 0, C_L = 50pF		0.2		V/ns
Falling edge slew rate	T_F	DRV = 0, C_L = 50pF		0.3		V/ns
Open-drain option						
Output low voltage	V_{OL}	DRV = 0 (I_O = 4mA)			0.4	V
Open drain leakage current					10	μA

GENERAL CHARACTERISTICS

V_{DD} = 3.3V or 5V, 30 mT ≤ B_Z ≤ 100 mT, T_A = -40°C to +150°C, typical values are at 25°C, unless otherwise noted.

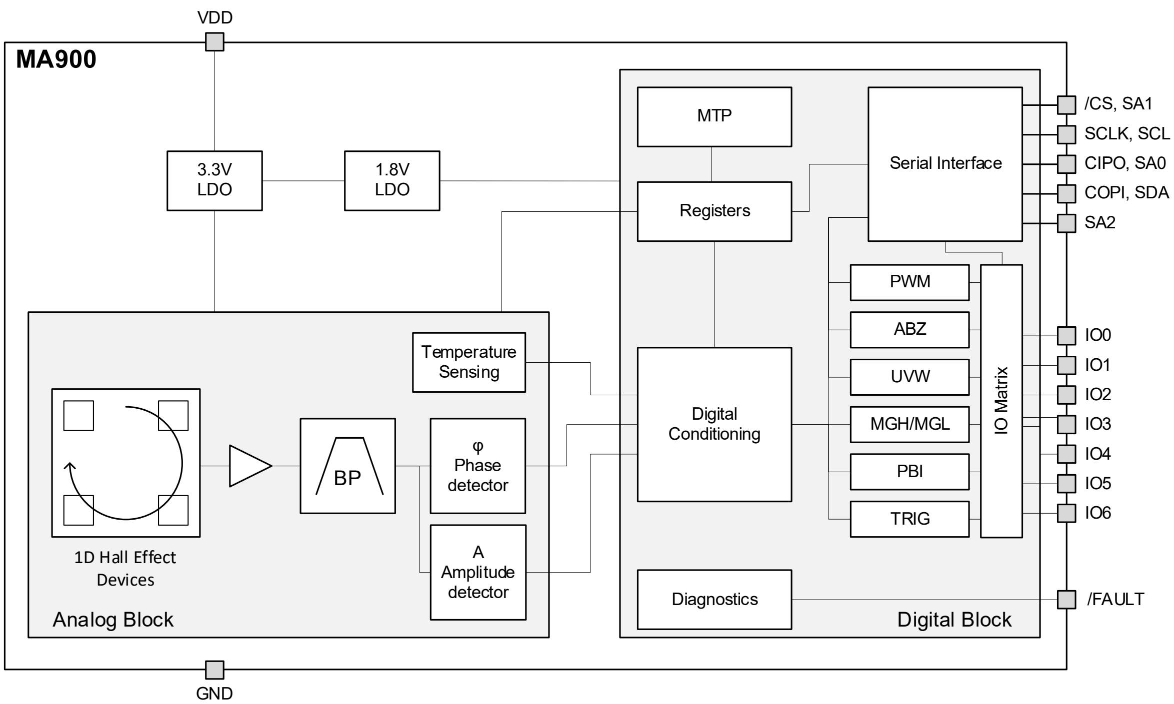
Parameter	Symbol	Condition	Min	Typ	Max	Units
Absolute Output – Serial outputs						
Data output length				16		bit
Resolution		FW = 8 (default)	TBD	12.4		bit
		FW = 13	TBD	14.5		bit
RMS Noise		FW = 8 (default)	TBD	0.01		deg
		FW = 13	TBD	0.003		deg
Refresh rate	$f_{REFRESH}$		970	1000	1030	kHz
Response time						
Power-up time		FW = 8 (default)			2	ms
Latency	L	FW = 0		9		µs
		FW > 0		0	1	µs
Filter cutoff frequency	f_{CUTOFF}	FW = 8 (default)		1560		Hz
Accuracy						
Angle Error	INL	T_A = 25°C		0.2	TBD	deg
Differential Non-Linearity	DNL			5		%
Output Drift						
Temperature induced drift		B_z = 40mT, T_A = 25°C		±0.01	TBD	deg/°C
Temperature induced variation		B_z = 40mT, T_A = -40°C to +85°C		±0.25	TBD	deg
		B_z = 40mT, T_A = 85°C to +150°C		±0.6	TBD	deg
Magnetic field induced		T_A = 25°C		±0.005	TBD	deg/mT
Voltage supply induced		B_z = 40mT, T_A = 25°C			±0.35	deg/V
External homogeneous field influence		B_z = 40mT, T_A = 25°C			0.03	°INL/mT
Absolute Output – PWM						
PWM Frequency	f_{PWM}	PWMF = 0	TBD	250	TBD	Hz
		PWMF = 1	TBD	1000	TBD	Hz
PWM Resolution				12		bit
Incremental Output – ABZ						
ABZ update rate				16		MHz
ABZ edges per turn			4		8192	
ABZ Pulses per channel per turn	PPT+1		1		2048	
ABZ hysteresis	H	Configurable	0.0055		15.9	deg
Overall ABZ jitter (peak- peak)		PPT = 511, speed = 1krpm		0.11		deg
Incremental Output – UVW						
UVW cycles per turn	NPP+1		1		8	
UVW hysteresis	H	Configurable	0.0055		15.9	deg
UVW jitter (3σ)		Speed = 1krpm		0.23		deg

GENERAL CHARACTERISTICS *(continued)*

V_{DD} = 3.3V or 5V, 30 mT ≤ B_Z ≤ 100 mT, T_A = -40°C to +150°C, typical values are at 25°C, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
Speed Output						
Speed Scale	S_{SPEED}	Range = ±100krpm		3.576		rpm/LSB
Speed output error		Speed = 10krpm			3	%
Speed Noise RMS		FWS = 7		125		rpm
Temperature output						
Temperature output sensitivity	S_T		TBD	8	TBD	LSB/°C
Magnetic Field Detection Thresholds						
Offset				3.5		mT
Gain Error			TBD	12	TBD	%
Temperature drift				±300		ppm/°C
MTP Memory Operations						
MTP Store time	t_{STORE}		650			ms
MTP Restore time	$t_{RESTORE}$		55			µs

BLOCK DIAGRAM



OPERATION

SENSOR FRONT-END

The magnetic field perpendicular to the die surface is detected with four integrated Hall devices located in the corners of the die, on a circle of 2.2 mm of diameter (Figure 1).

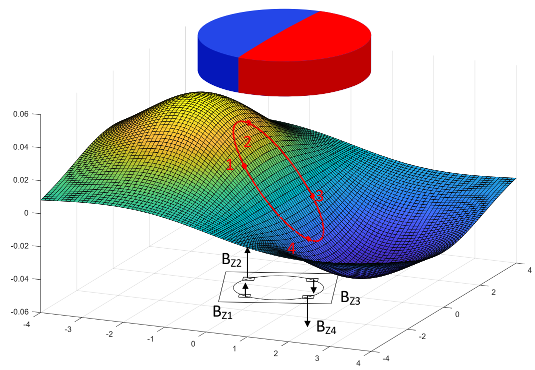
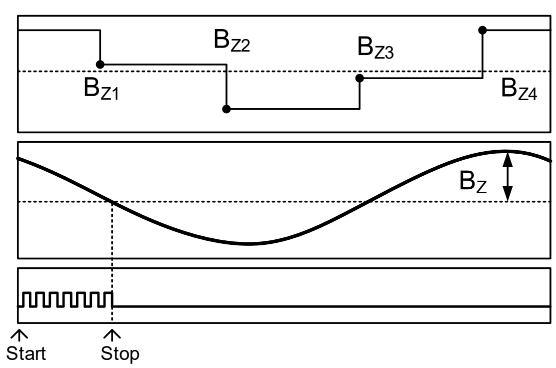


Figure 1: Hall devices positioning.

The four Hall devices are sequentially probed (Figure 2 – Top) and a band pass filter centered on the measurement cycle frequency extracts the fundamental harmonic of the resulting signal (Figure 2– Mid). The phase of the filtered signal is proportional to the angle of the magnet and is not affected by external homogeneous (stray) fields. The digital angle is obtained through a time-to-digital transformation (Figure 2 – Bottom). Digital data are then conditioned by a unit containing the digital filtering and the zero setting. This measurement technique does not require AD converters and ATAN calculation.



Top: Magnetic Field Samples
Mid: Fundamental Harmonic
Bottom: Clock of Time-to-Digital Converter
Figure 2:Phase Detection Method

Note: the quantity B_Z shown in Figure 2 is essentially the gradient of the magnetic field at the sensor center multiplied by the radius of the circle where the Hall devices are located (1.1 mm). In this document we refer to B_Z, the field amplitude in T, instead of the actual gradient in T/m, because B_Z is the quantity actually measured by the sensor and thus the one determining the sensor performances.

SENSOR – MAGNET MOUNTING

The differential measurement technique used in MA900 requires an end-of-shaft configuration (see Figure 3).

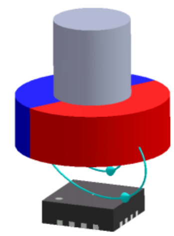


Figure 3: End-of-Shaft Mounting

In such configuration, the exact shape and regularity of the magnet do not matter: when the sensor is placed on the magnet rotation axis the phase of the fundamental harmonic will always be proportional to the magnet angle

A typical magnet is a Neodymium alloy (N35) cylinder with 6 mm diameter and 2.5 mm height, inserted into an aluminum shaft. The typical air gap between magnet and sensor is between 1 mm and 2 mm (Figure 4). For good measurement linearity, the sensor shall be positioned with a radial precision of typically 0.5 mm.

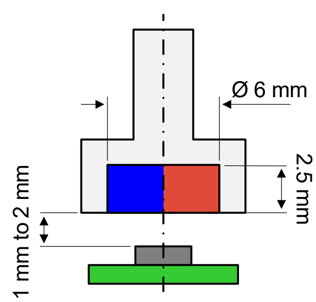


Figure 4: Typical magnet dimension and placement (cross-section)

Magnets with either diametral or bi-axial magnetization (see Figure 5) are suitable for use with MA900. The two magnetization options have complementary benefits and drawbacks.

For equal magnet material, dimension, and sensor distance:

- Diametral magnet generates a smaller field , but is more tolerant to mounting imperfections (e.g. sensor/magnet misalignment)
- Bi-axial magnet generates a larger field , but the measurement accuracy will be more affected by mounting imperfections .

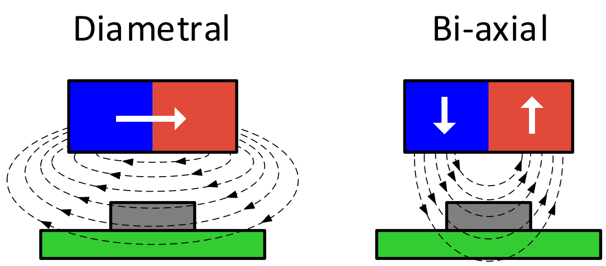


Figure 5: Diametral and Bi-axial magnetization

By default, when looking at the package top, the angle increases when the shaft rotates clockwise. Figure 6 shows the zero angle of the unprogrammed sensor and the location of the sensitive points. Both the rotation direction and the zero angle can be programmed.

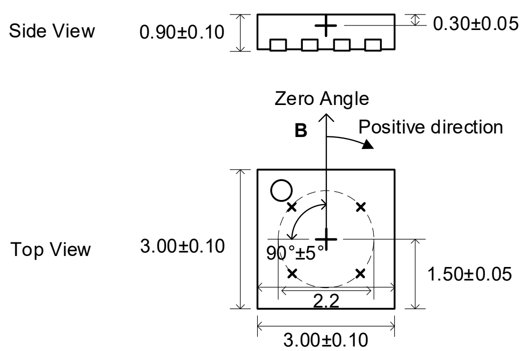


Figure 6: Detection Points and Default Positive Direction

ELECTRICAL MOUNTING AND POWER SUPPLY DECOUPLING

For both 5V and 3.3V operation, it is recommended to place a 1µF decoupling capacitor, connected to VDD pin close to the sensor with a low impedance path to GND pin (Figure 7).

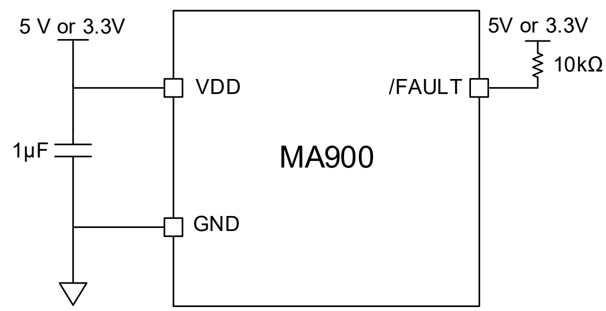


Figure 7: Connection for supply decoupling.

SERIAL INTERFACES

SPI

SPI is a four-wire, synchronous, serial communication interface. The MA900 supports SPI Mode 0 and Mode 3 (see Table 1 and Table 2). The SPI Mode is detected automatically by the sensor and therefore does not require any action from the user. The maximum SPI clock frequency supported by MA900 is 20MHz, and there is no minimum clock rate. Note that actual data rates depend on the PCB layout quality and signal trace length. See Figure 8, Figure 9 and Table 3 for details on SPI timing.

In this document, COPI stands for Controller Output Peripheral Input, CIPO stands for Controller Input Peripheral Output.

All commands to the MA900 (whether for writing or reading register content) must be transferred through the SPI COPI pin, see the SPI Commands section for details.

Table 1: SPI Specification

	Mode 0	Mode 3
SCLK Idle State	Low	High
Data Capture	On SCLK rising edge	
Data Transmission	On SCLK falling edge	
/CS Idle State	High	
Data Order	MSB first	

Table 2: SPI Standard

	Mode 0	Mode 3
CPOL	0	1
CPHA	0	1
Data Order (DORD)	0 (MSB first)	

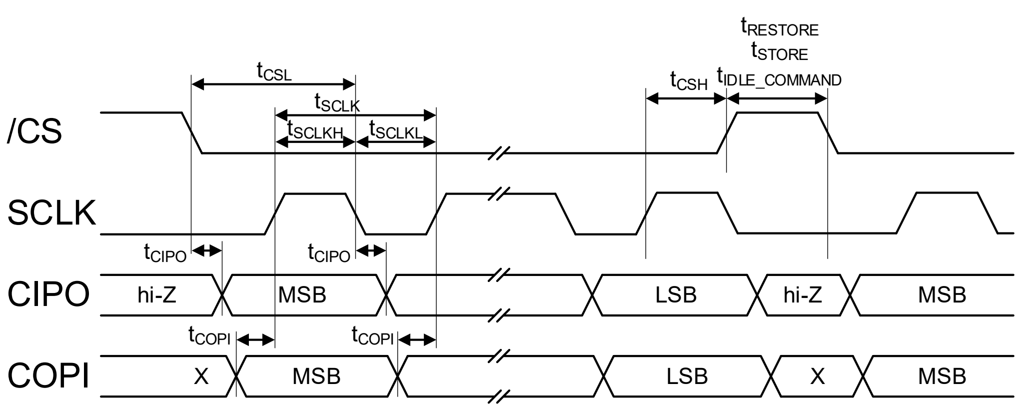


Figure 8: SPI Mode 0

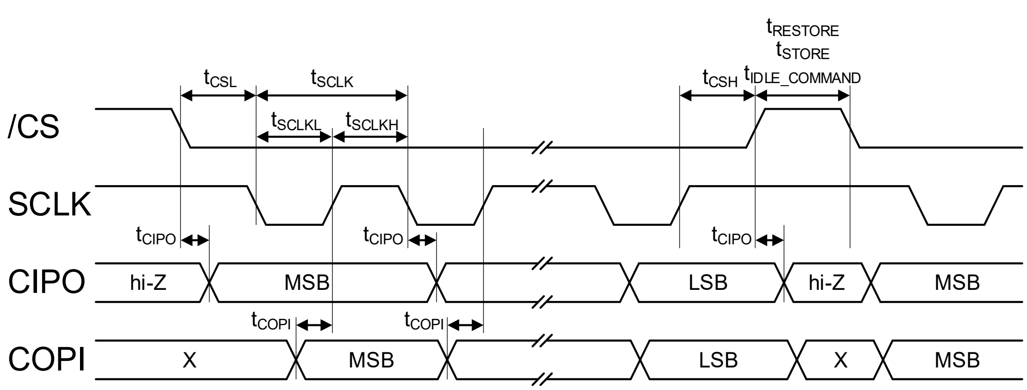


Figure 9: SPI Mode 3

Table 3: SPI Timings

Parameter [8]	Description	Min	Max	Unit
$t_{IDLE\_COMMAND}$	Idle time between transmissions	100		ns
t_{CSL}	Time between /CS falling edge and SCLK falling edge	25		ns
t_{SCLK}	SCLK period	50		ns
t_{SCLKL}	Low level of SCLK signal	25		ns
t_{SCLKH}	High level of SCLK signal	25		ns
t_{CSH}	Time between SCLK rising edge and /CS rising edge	25		ns
t_{CIPO}	SCLK setting edge to data output valid		20	ns
t_{COPI}	Data input valid to SCLK reading edge	20		ns

SPI Signals Routing on PCB

Typically, a direct connection between the sensor and controller SPI pins is sufficient to ensure a proper data transfer over the SPI bus. However, in specific situations (e.g.: long signal traces, external EMI presence) it is possible to improve signal integrity by making special considerations during the PCB design in particular for SCLK line. A list of useful guidelines is shown below:

- Properly shield all SPI signals with a GND plane on both sides of each trace, as well as GND plane underneath the SPI signals.
- Place vias along these traces to connect the top and bottom GND planes.
- To avoid EMI issues, route the SCLK signal away from the other SPI signals and noise sources. The distance should be at least three times the SCLK trace width.
- Insert an RC low-pass filter on SCLK (see Figure 10). This RC filter must be located close to the sensor. It is recommended to use a 200Ω serial resistor with a 10pF shunt capacitor to obtain a 80MHz filter cutting frequency.
- Use a star topology for the GND connection and keep it as direct and short as possible to avoid ground loops.
- Insert RC low-pass filters on the CIPO and COPI signals (see Figure 10). The RC filter on COPI must be located close to the controller, while the filter on CIPO must be located close to the sensor. It is recommended to use a 200Ω resistor with a 10pF capacitor.
- Avoid a significant trace length mismatch between the SPI signals, especially between the CIPO, COPI, and SCLK signals. These signals should have similar propagation delays.
- If possible, avoid placing vias on the SCLK signal.

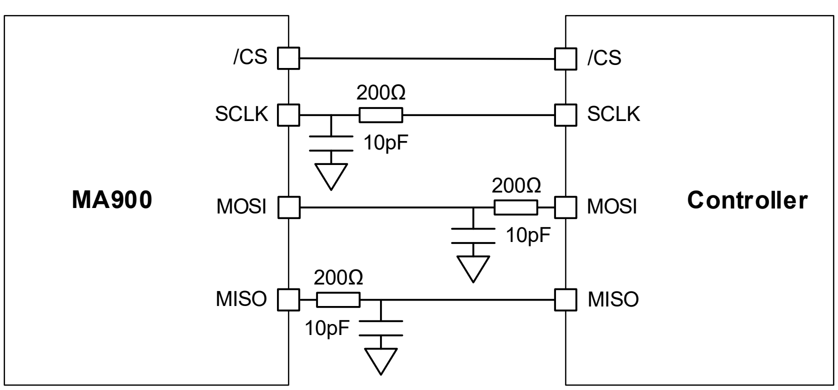


Figure 10: Examples of RC Low-Pass Filters on the SPI Signals

SPI Commands

MA900 responds to two SPI commands:

- Read (one or multiple consecutive registers)
- Write (one or multiple consecutive registers)

Angle, Speed and Multiturn measurements can be obtained by sending a Read command on specific registers.

MTP Store/Restore operations are executed by sending specific Write commands on register 120 (see MTP Operations section).

SPI Read and Write commands always return the 16-bit angle at the beginning of command, allowing to minimize the measurement lag.

There are two mechanisms for communication error detection:

- Data transmitted on COPI/CIPO can be protected with a 4-bit CRC code (CRC4) for every 12 bits of data. See APPENDIX A: CRC CALCULATION for CRC calculation details.
- A Handshaking protocol can be activated for the transmitter to verify that the receiver got the correct data.

In case of system failure detection, the SPI communication can be intentionally corrupted by inverting the CRC.

SPI Commands Common Features

The SPI frames used for both Read and Write commands share some common characteristics:

- The communication is organized in 16-bit sized frames.
- The controller must send an operation code (OpCode) on the first 4 bits (MSBs) of the first 16-bit frame on COPI. This code is used to distinguish between read or write command.
- In the following 8 bits, the controller must send the address ($Addr_N$) of the register on which the read or write command shall be performed.
- The last 4 bits of each 16-bit frame are reserved for the CRC4, which is calculated on the value of the 12 preceding bits. If CRC is enabled, the controller shall send a correct CRC4 to avoid a communication error. The sensor always sends the CRC4 regardless of the value of CRCEN.
- The 16-bit angle value is sent by the sensor in the first 12 bits of the first frame ($Angle_{MSB}$) and the first 4 bits of the second frame ($Angle_{LSB}$) on CIPO. The 16-bit value is obtained as $Angle_{MSB}$*16 + $Angle_{LSB}$.
- If SPI Handshaking is enabled (SPIHS = 1) and a SPI Read is being performed, starting from the second 16-bit frame the controller must send on COPI the 16-bits received on CIPO in the previous frame.
- Regardless of SPI Handshaking status, when a SPI Write is being performed, the sensor sends on CIPO the 16-bits received on COPI in the previous frame (starting from the second 16-bit frame).

SPI Read

The SPI Read command allows to read a single register or multiple consecutive registers. The format for a single register read is shown in Figure 11:

- The OpCode to use for SPI Read commands is 0b0011
- In the second 16-bit CIPO frame, $Angle_{LSB}$ is followed by the 8-bit $Data_N$, namely the content of register $Addr_N$.

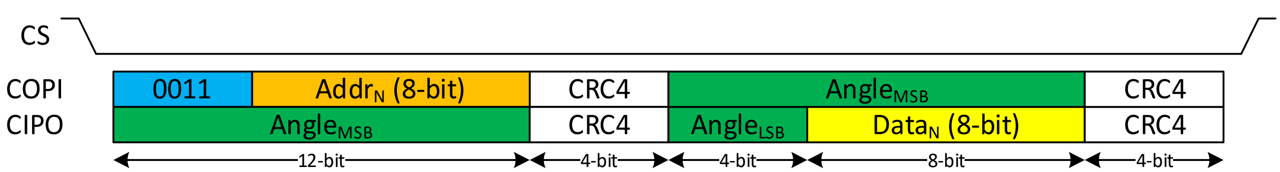


Figure 11: SPI Read Single

The SPI Read command can be extended to read a sequence of consecutive registers (Read Burst).

In each additional 16-bit frame *i*, the sensor sends the value of the register at $Addr_{N+i}$ on

CIPO. The first 4 bits of each frame always contain $Angle_{LSB}$.

The format for a 2-registers Read Burst is shown in Figure 12

The length of SPI Read Burst is limited only by the number of available registers to read.

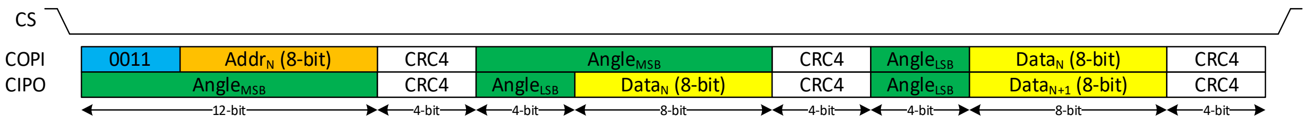


Figure 12: SPI Read Burst

SPI Write

The SPI Write command allows to write a single register or multiple consecutive registers. The format for a single register write is shown in Figure 13:

- The OpCode to use for SPI Write commands is 0b1100.
- In the second 16-bit COPI frame, the controller shall send 0b000 followed by the 8-bit data ($Data_N$) to write in register $Addr_N$.
- The 0b111100000000 code is transmitted via COPI to indicate the end of SPI Write. All additional data transmitted on COPI after this code are ignored.

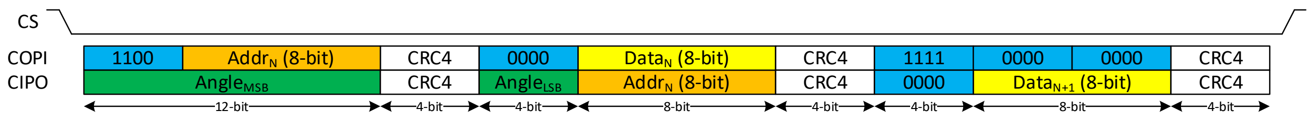


Figure 13: SPI Write Single

The SPI Write command can be extended to read a sequence of consecutive registers (Write Burst).

In each additional 16-bit frame *i*, the controller must send on COPI the value to write in the register at $Addr_{N+i}$.

As for single write, the controller must send via COPI the 0b111100000000 code after the last write value.

The SPI Write format for a 2-registers Write Burst is shown in Figure 14

The length of SPI write burst is limited only by the number of available registers to write.

If CRC is enabled (CRCEN = 1), the register value is changed only if the CRC4 sent by the controller is correct. If an invalid CRC4 is detected, the SPI protocol error flag is raised, the current write command and all the following ones are ignored.

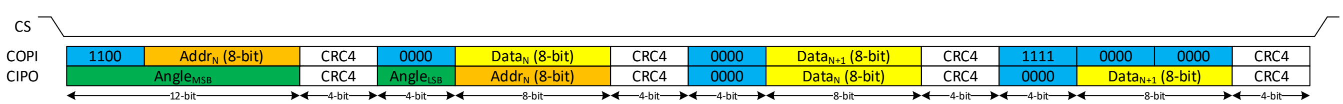


Figure 14: SPI Write Burst

SPI Read Angle

Any SPI Read or SPI Write command can be used to read the 16-bit Angle, since this value is always sent by the sensor on COPI in the first two frames of each command. Anyway, it is recommended to read the angle via a SPI Read of Register 107 (Figure 15). Register 107 value is an 8-bit counter which is incremented each time the register is read. This allows the controller verify that the sensor is working properly by checking if the counter value is updated at every SPI Read.

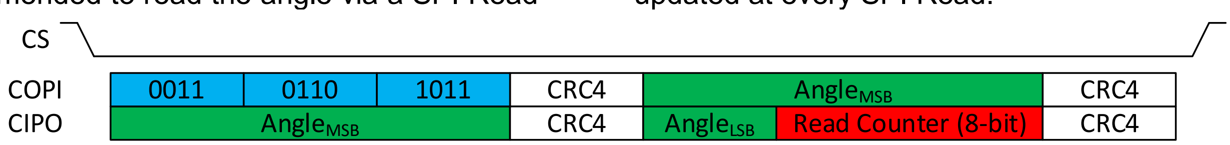


Figure 15: SPI Read Angle (Register 107)

There are several options to perform shorter angle reads for time optimization:

- It is possible to read a 16-bit Angle with 20 SPI clock cycles (Figure 16). In this case, though, the 4-bit $Angle_{LSB}$ is not protected by CRC and an eventual communication error will not be detected.

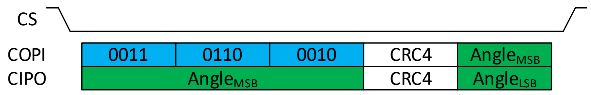


Figure 16: 20 clock cycles Read Angle

- It is possible to read a 12-bit Angle and the CRC4 with 16 SPI clock cycles (Figure 17)

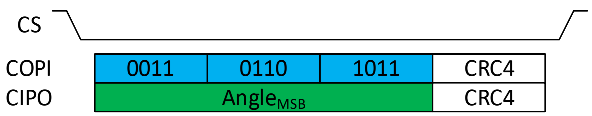


Figure 17: 16 clock cycles Read Angle

- It is possible to read a 12-bit Angle without CRC4 in 12 clock cycles (Figure 18). In this case too, an eventual communication error will not be detected.

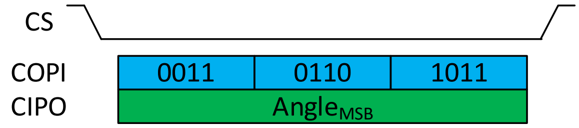


Figure 18: 12 clock cycles Read Angle

SPI Error Detection

The MA900 can detect several types of errors during the SPI communication. Whenever an error is detected, the /FAULT pin is driven low, the SPI error flag in Register 111 is set to 1 and the operation is converted to a Read of Register 108. If SPI CRC Inversion is enabled (CRCINVEN = 1), each CRC sent on CIPO after the error detection is bitwise inverted until the SPI error flag is cleared.

First Frame Invalid OpCode or CRC

An error is detected if the OpCode sent by the controller does not match either a Read (0b0011) or Write (0b1100) command (Figure 19).

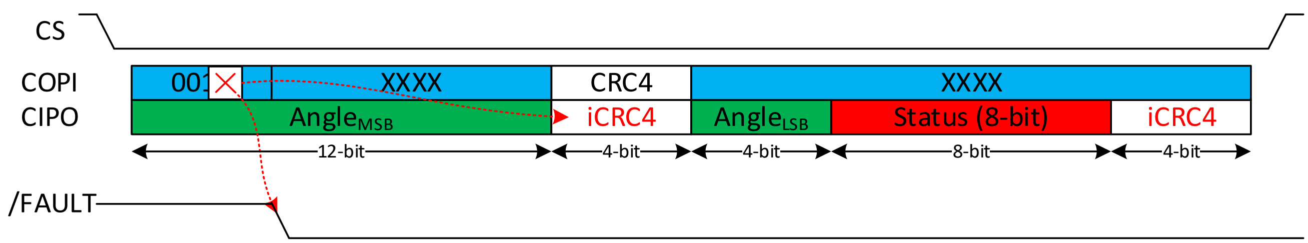


Figure 19: Wrong OpCode detection

A corruption of the address value can be detected if CRC is enabled. In this case, the error is detected if the first CRC4 received on COPI does not match the expected one (Figure 20 and Figure 21).

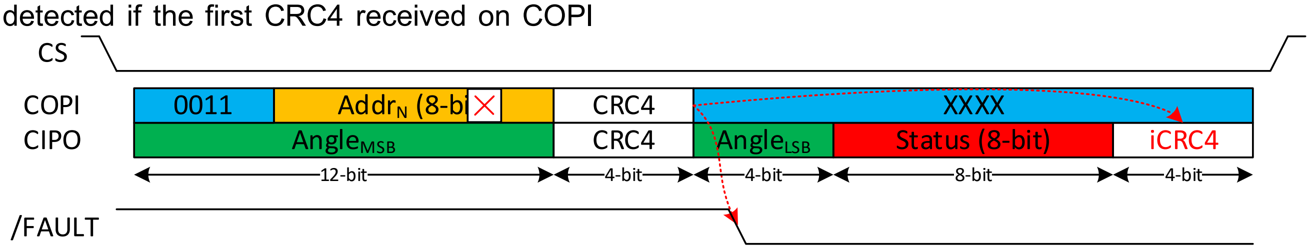


Figure 20: Error detection after valid Read OpCode

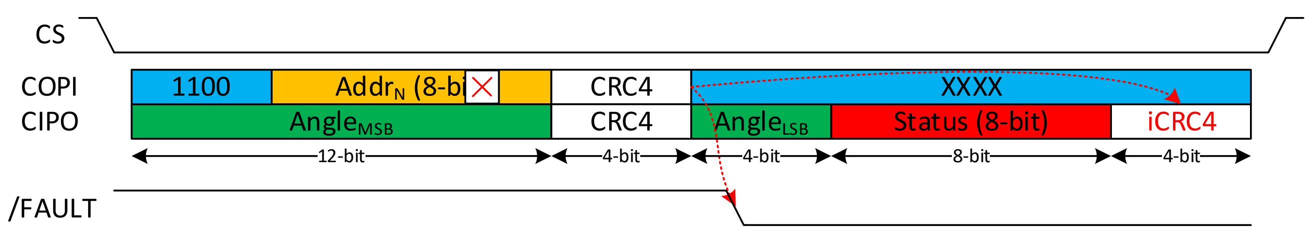


Figure 21: Error detection after valid Write OpCode

Write Error

Data corruption during a Write Command is detected when CRC is enabled, as the CRC4 sent after the data does not match the expected value (see $Data_{N+1}$ in Figure 22).

In this case, the write is not performed. If Handshaking is enabled (SPIHS = 1), the controller can detect this by checking that the data returned by the sensor does not match the sent data.

It is recommended to end the Write operation with 0b111100000000 code to verify that the last write operation was performed correctly.

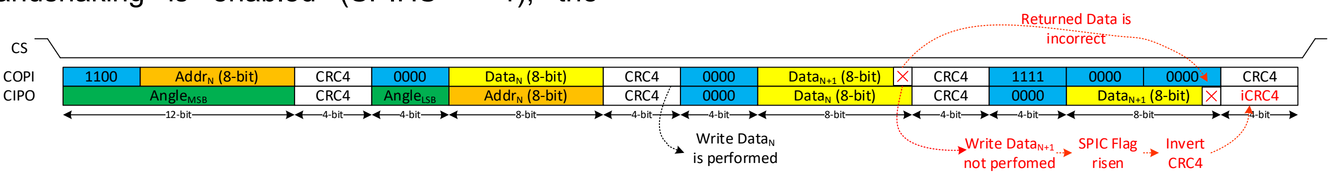


Figure 22: SPI Write Error

I²C

The I²C implementation used in MA900 follows the I²C standard for bidirectional, single controller communication. The sensor operates only as a peripheral device and is designed to operate in the Fast-Mode Plus (FM+) I²C speed category (1Mbit/s).

MA900 I²C communication can be secured by CRC. When CRC is enabled (CRCEN = 1) an 8-bit CRC code (CRC8) is transmitted after each byte read or write. See APPENDIX A: CRC CALCULATION for CRC calculation details.

Mandatory I²C features for peripheral device defined by UM10204 (NXP I²C-bus specification and user manual) are supported.

I²C Commands

The following I²C commands are supported on MA900:

- Read
- Quick Read
- Write

Each command can be performed on a single or multiple consecutive registers.

I²C Read

An I²C Read operation allows the controller to receive the value of one or multiple consecutive registers. It is formed by the following sequence:

- The controller shall send a start condition, followed by 8 bits. The first 7 bits shall contain the Sensor Address, the last bit shall be 0.
- If the Sensor Address is recognized, the sensor sends an acknowledge (ACK) bit, pulling the SDA line to 0.
- The controller shall send 8 bits containing the address of the register to be read.
- The sensor sends the ACK bit.
- The controller shall send a repeated start condition, followed by 8 bits. The first 7 bits shall contain the Sensor Address and the least significant bit shall be 1.
- The sensor sends an ACK bit, followed by the register 8-bit value.
- To stop the communication, the controller shall send a not-acknowledge bit (NACK) bit, by pulling the SDA line to 1, followed by a stop condition (Figure 23).
- To receive the value of the next register, the controller shall send an ACK bit and keep sending clock pulses on the SCL line (Figure 24). This operation can be repeated multiple times and is only limited by the number of available registers.
- If CRC is enabled (CRCEN = 1), the sensor sends the CRC8 value after each register value (Figure 25).

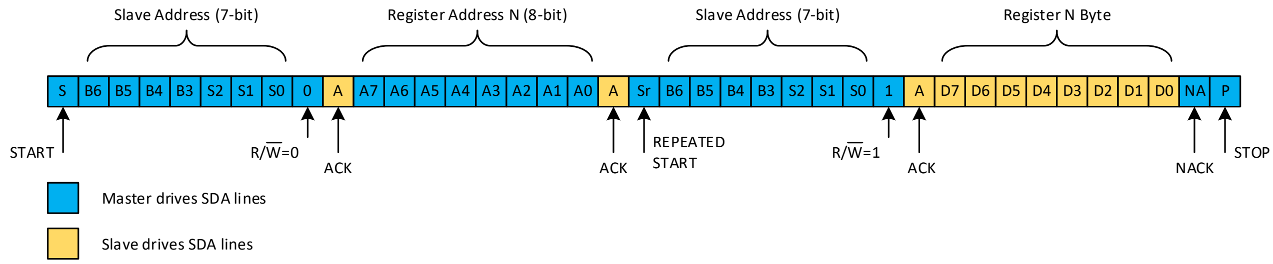


Figure 23: I²C Read Single

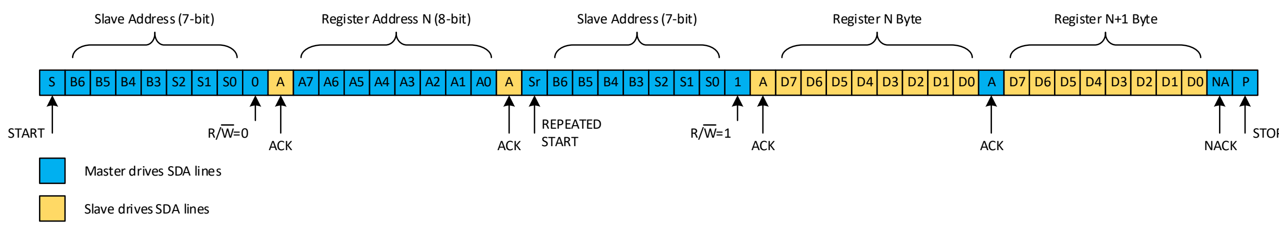


Figure 24: I²C Read Burst

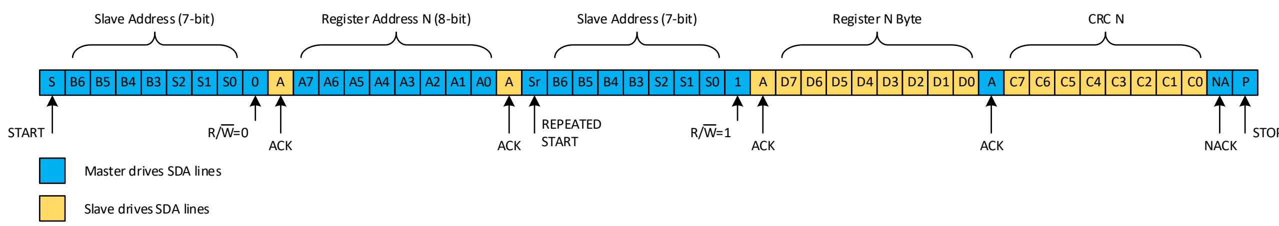


Figure 25: I²C Read Single with CRCEN = 1

I²C Quick Read

The Quick Read command is a faster option to read the measurement related registers (96 to 101). It is formed by the following sequence:

- The controller shall send a start condition, followed by 8 bits. The first 7 bits shall contain the Sensor Address and the least significant bit shall be 1.
- If the Sensor Address is recognized, the sensor sends an ACK bit, followed the register 96 ($ANGLE_{LSB}$) 8-bit value.
- As in the I²C Read case, the controller determines how many register values are sent by the sensor. Figure 26 shows an I²C Quick Read operation to read the 16-bit angle.
- If CRC is enabled (CRCEN = 1), the sensor sends the CRC8 value after each register value (Figure 27).

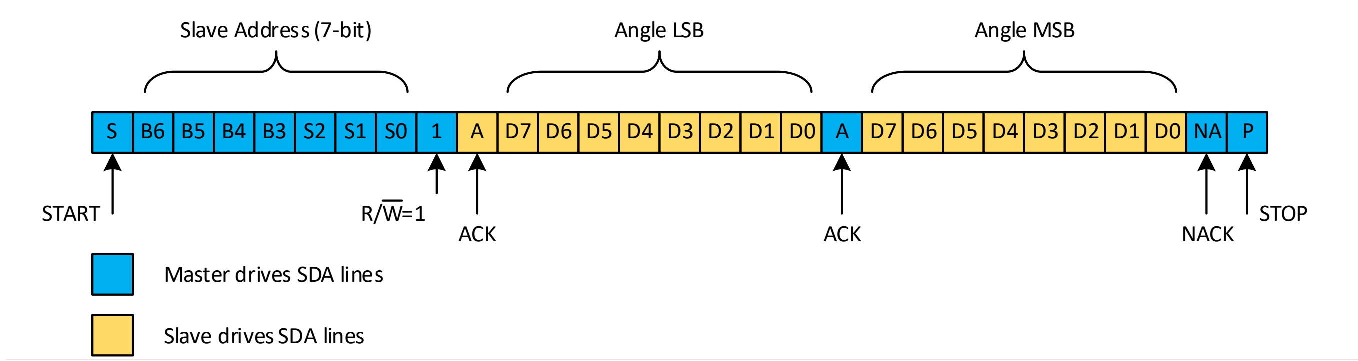


Figure 26: 16-bit angle I²C Quick Read

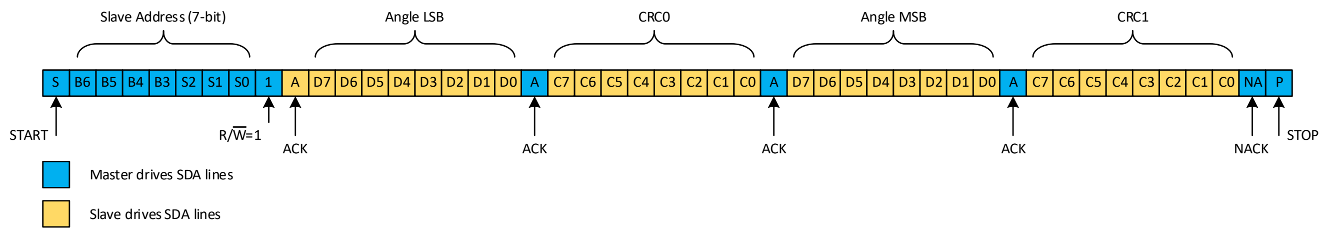


Figure 27: 16-bit angle I²C Quick Read with CRCEN = 1

I²C Write

The I2C Write command allows the controller to change the value of one or multiple consecutive registers. It is formed by the following sequence:

- The controller shall send a start condition, followed by 8 bits. The first 7 bits shall contain the Sensor Address and the last bit shall be 0.
- The sensor sends an acknowledge (ACK) bit if the Sensor Address is recognized.
- The controller shall send 8 bits containing the address of the register where the new value needs to be written.
- The sensor sends the ACK bit.
- The controller shall send the 8-bit value to write in the register.
- The sensor sends the ACK bit.
- The stop the communication, the controller shall send a stop condition (Figure 28)
- To write in the following register, the controller shall send the 8 bit value to write in it (Figure 29).
- If CRC is enabled (CRCEN = 1), the controller shall send the CRC after each value sent (Figure 30). If the controller does not send the CRC or sends a wrong CRC, the I²C Write is ignored and the IFAIL (Register 108) and I²C (Register 111) flags are set to 1.

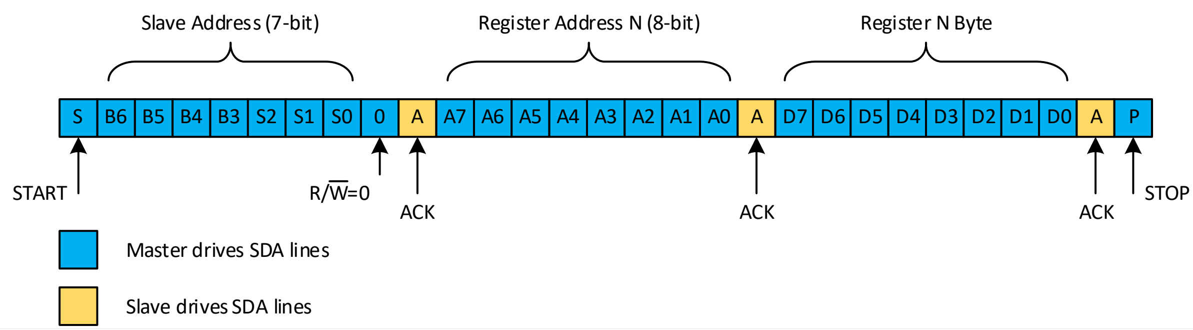


Figure 28: I²C Write Single

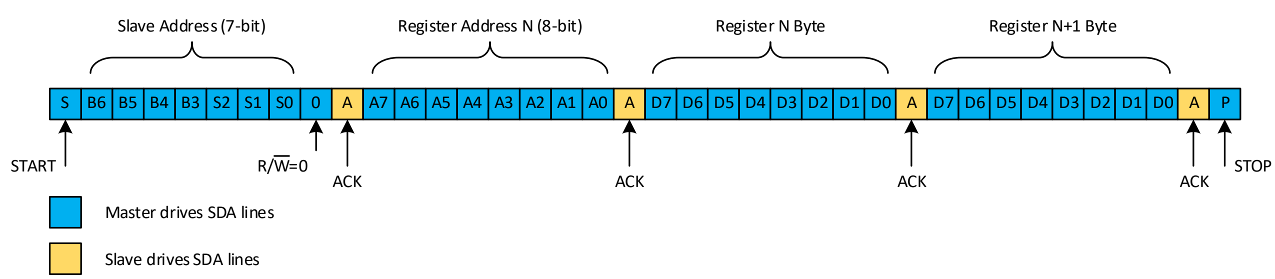


Figure 29: I²C Write Burst

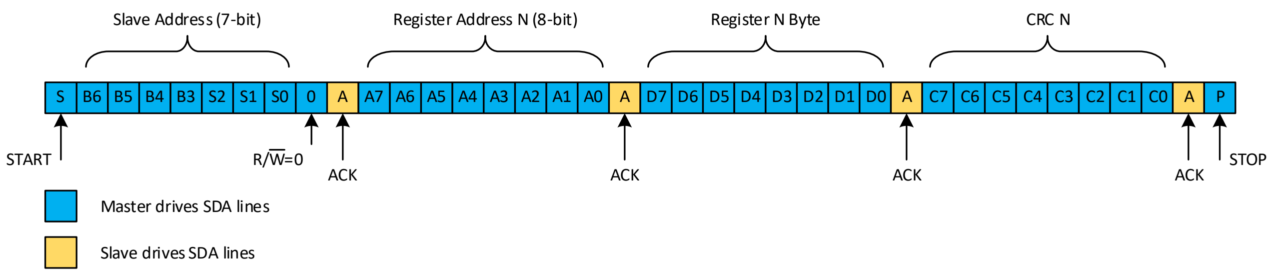


Figure 30: I²C Write with CRC

Measurement values latching

By default, ANGLE/SPEED/MTV/TEMP values are latched during the entire I²C transactions, namely between a start and a stop condition. This ensures all the values read during the I²C transactions are coherent.

SPI/I²C CONFIGURATION

Interface Selection

The serial interface used by MA900 is determined by the value of I2CEN bit (see Table 4). The interface selection is happening only at start-up, thus any modification to I2CEN bit does not switch the interface immediately.

Table 4: Serial Interface Selection

I2CEN	Serial Interface
0	SPI
1	I²C

I2CEN register can only be modified if MTPLOCK=0 and WRLOCK=0 (see Lock Mechanisms section).

To switch from SPI to I²C or vice versa, the user shall then:

1. Disable MTP Lock and Write Lock.
2. Change I2CEN bit value.
3. Perform an MTP Store (see MTP Store section).
4. Power cycle the MA900.

I²C Sensor Address Selection

The I²C 7-bit sensor address is defined by the logic level of SA0, SA1, SA2 pins at start-up (see Table 5). The sensor address is not updated if SA0, SA1, SA2 pins value changes after startup.

Table 5: I²C Sensor Address

I²C Sensor Address (7-bit)						
1	1	0	0	SA2	SA1	SA0

CRC

The CRC, for both SPI and I²C, can be enabled or disabled via the CRCEN bit (Table 6).

Table 6: CRCEN Setting

CRCEN	CRC Status
0	Disabled
1 (default)	Enabled

When the CRC is disabled, the sensor does not check if the CRC4/CRC8 value provided by the controller is correct. Anyway, if SPI is used, the sensor continues to calculate and transmit the CRC4 on CIPO line.

SPI CRC Inversion

It is possible to enable or disable the SPI CRC inversion via the CRCINVEN bit (Table 7).

Table 7: CRC Inversion Setting

CRCINVEN	CRC sent on CIPO after error detection
0	Normal
1 (default)	Inverted

When SPI CRC inversion is disabled (CRCINVEN = 0), the sensor sends a normal CRC4 even if an error is detected (see Figure 31)

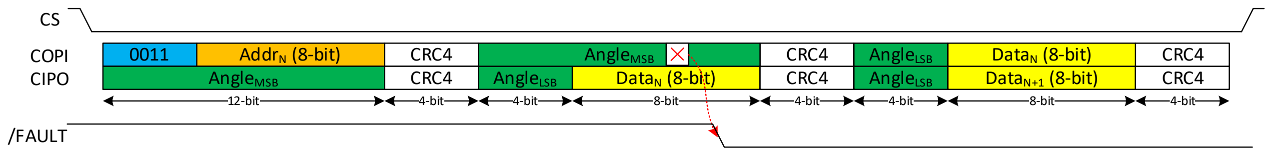


Figure 31: CRC4 sent on CIPO when CRCINVEN = 0

SPI Handshaking

The SPI Handshaking can be enabled or disabled via the SPIHS bit (Table 8):

Table 8: Handshaking Protocol Setting

SPIHS	Handshaking protocol
0 (default)	Disabled
1	Enabled

When the SPI Handshaking is disabled, the controller does not need to send back on COPI the data received in the previous frame on CIPO.

If CRC is enabled, though, the CRC4 sent by the controller, still needs to be valid.

Figure 32 shows a Read Angle operation with Handshaking disabled.

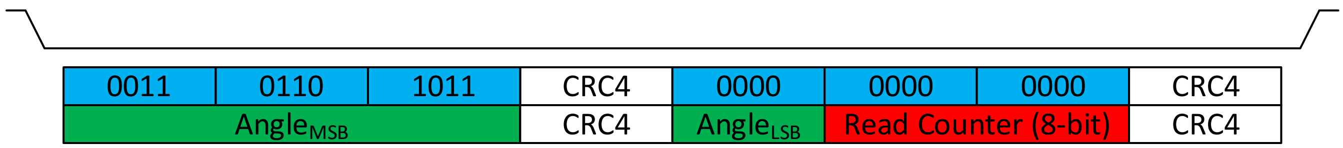


Figure 32: Read Angle without Handshaking

MTP MEMORY OPERATIONS

Register 0 to 31 values can be stored in MTP (multi-time programmable) memory, to keep the sensor settings after a power cycle. MTP Store/Restore operations can be performed both via SPI and I²C.

MTP Store

MTP Store operation allows to save the current values of registers 0 to 31 in the MTP.

This operation is performed by setting the STORE bit in register 120 to 1. The bit is automatically cleared once the operation is completed. The time required to perform an MTP Store is shown in the General Characteristics table.

MTP Restore

MTP Restore operation allows to copy the values stored in the MTP to registers 0 to 31. This operation is performed by setting the RESTORE bit in register 120 to 1. The bit is automatically cleared once the operation is completed. The time required to perform an MTP Restore is shown in the General Characteristics table.

MTP Flags

During MTP operations, the MTPBUSY flag in Register 108 is set to 1 and MTPDONE flag in Register 120 is set to 0. When MTPBUSY = 1, registers 0 to 31 are not accessible. (neither Read nor Write operations are allowed). If a register Read or Write operation is performed on any of these registers, the operation is discarded, the IFB flag of IFAIL is set and /FAULT pin is driven low.

After a successful MTP operation, MTPDONE flag is set to 1.

SSI

SSI is a 2-wire synchronous serial interface. The sensor operates as a peripheral to the external SSI controller.

SSI can only be used for angle reading. It is not possible to read or write registers using SSI.

The maximum SSI clock frequency supported by MA900 is 5MHz. The actual data rates depend on the PCB layout quality and signal trace length.

Two SSI modes are supported:

- In Mode A, SSCK is held high when communication is idle (Figure 33)
- In Mode B, SSCK is held low when communication is idle (Figure 34)

Timing details of SSI communication are shown in Table 9.

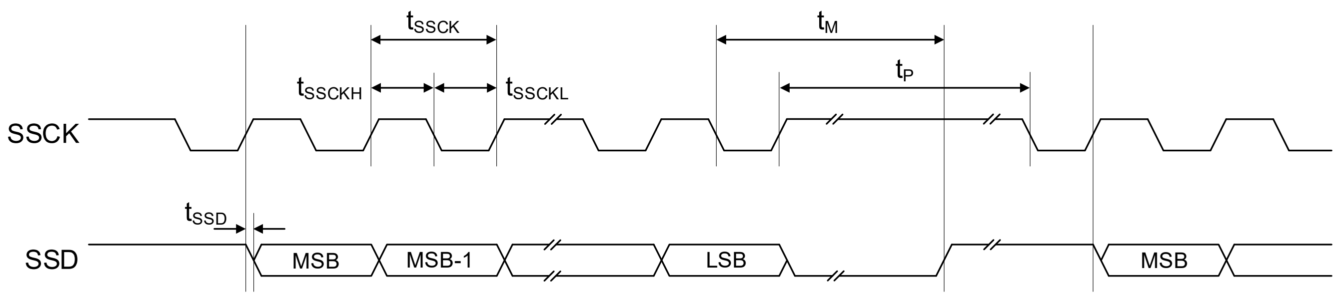


Figure 33: SSI Mode A

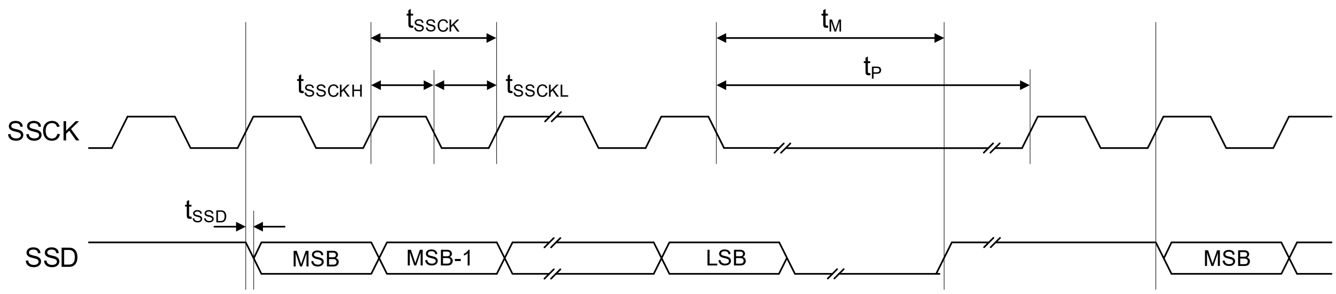


Figure 34: SSI Mode B

Table 9: SSI Timings

Parameter	Description	Min	Max	Unit
t_{SSD}	Delay between SSCK and SSD edge		50	ns
t_{SSCK}	SSCK period	0.2	20	µs
t_{SSCKL}	Low level of SSCK signal	0.1	10	µs
t_{SSCKH}	High level of SSCK signal	0.1	10	µs
t_M	Transfer timeout (monoflop time)	25		µs
t_P	Dead time: SSCK high time for next data reading	40		µs

SSI Read Angle

To read the angle via SSI, the controller shall send clock pulses on SCLK. The first edge on SSCK (falling for Mode A, rising for Mode B) is the trigger event signaling the start of a new transmission. After the first edge detection, the sensor sends on SSD the angle 16-bit sequence, MSB first. The SSD level is updated on each SSCK rising edge and the controller shall read it on the following falling edge.

Since the first clock pulse on SSCK is used to trigger a new transmission, reading the 16-bit angle requires 17 clock pulses (see Figure 35 and Figure 36). The transmission length can be reduced if a shorter angle data is sufficient for the application. For example, if an application requires only a 12-bit angle information, it is sufficient to send 13 clock pulses on SSCK.

When a trigger event is detected, the current angle data is held in the output buffer until the transfer timeout t_M has passed from the last SSCK falling edge (see Table 10).

Consecutive angle readings shall be separated by a minimum delay t_P see (Figure 37).

Table 10: Sensor Data Timing

Trigger Event	Release of the Output Buffer
First SSCK rising edge after dummy clock	Last SSCK falling edge + time out t_M

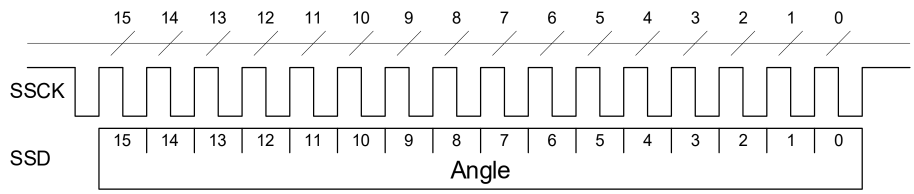


Figure 35: SSI Mode A Angle Read

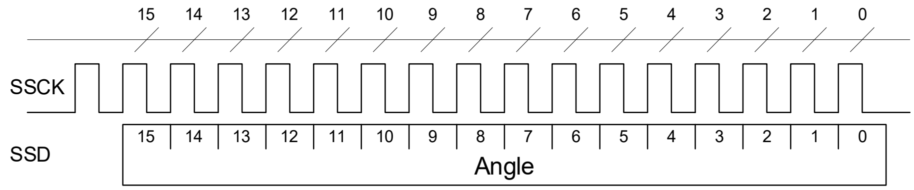


Figure 36: SSI Mode B Angle Read

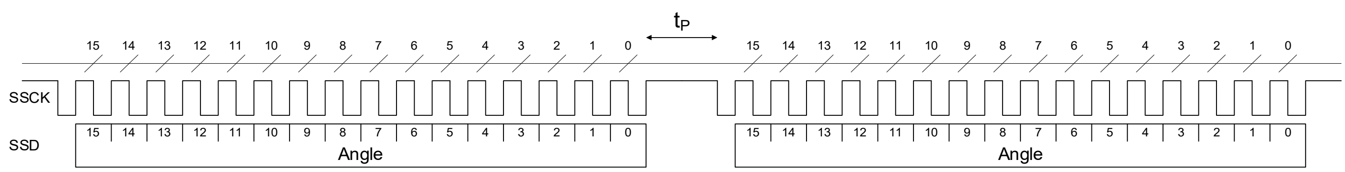


Figure 37: Two consecutive 16-bit SSI Angle Reads (Mode A)

SSI Parity Check

SSI data integrity can be validated via a parity bit.

The parity bit is always sent after the angle LSB. To read the parity bit, then, the controller needs to send 18 clock pulses (Figure 38).

The parity sign can be configured via the SSIP bit in Register 12 (Table 11).

Table 11: SSI Parity Sign

SSIP	SSI Parity Sign
0 (default)	Even
1	Odd

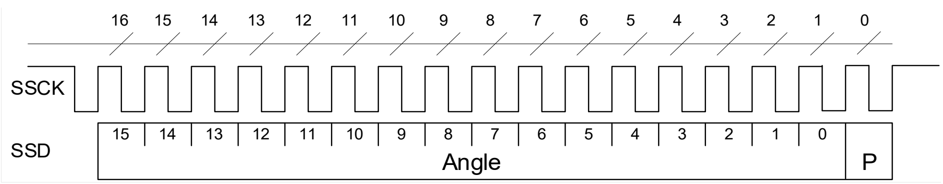


Figure 38: SSI Angle Read with Parity Check

REGISTER MAP

Table 12: Register Map

<table>
<tr><th>Add</th><th>Page</th><th>Hex</th><th>R/W</th><th>Bit 7</th><th>Bit 6</th><th>Bit 5</th><th>Bit 4</th><th>Bit 3</th><th>Bit 2</th><th>Bit 1</th><th>Bit 0</th></tr>
<tr><td>0</td><td rowspan="20">0</td><td>00</td><td>RW</td><td colspan="8">ZERO[7:0]</td></tr>
<tr><td>1</td><td>01</td><td>RW</td><td colspan="8">ZERO[15:8]</td></tr>
<tr><td>2</td><td>02</td><td>RW</td><td colspan="8">HYST[7:0]</td></tr>
<tr><td>3</td><td>03</td><td>RW</td><td colspan="8">PPT[7:0]</td></tr>
<tr><td>4</td><td>04</td><td>RW</td><td>DAZ</td><td colspan="4">ILIP[3:0]</td><td colspan="3">PPT[10:8]</td></tr>
<tr><td>5</td><td>05</td><td>RW</td><td colspan="5">-</td><td colspan="3">NPP[2:0]</td></tr>
<tr><td>6</td><td>06</td><td>RW</td><td colspan="4">MGLT[3:0]</td><td colspan="4">MGHT[3:0]</td></tr>
<tr><td>7</td><td>07</td><td>RW</td><td>-</td><td colspan="3">PBI_TIME[2:0]</td><td>PBI_EN</td><td colspan="3">RESERVED</td></tr>
<tr><td>11</td><td>0B</td><td>RW</td><td colspan="6">-</td><td>PWMM</td><td>PWMF</td></tr>
<tr><td>12</td><td>0C</td><td>RW</td><td colspan="7">-</td><td>SSIP</td></tr>
<tr><td>15</td><td>0F</td><td>RW</td><td>RWM</td><td>I2CEN</td><td>-</td><td>WLOCK</td><td>-</td><td>SPIHS</td><td>CRCINVEN</td><td>CRCEN</td></tr>
<tr><td>16</td><td>10</td><td>RW</td><td colspan="7">-</td><td>RD</td></tr>
<tr><td>17</td><td>11</td><td>RW</td><td colspan="4">-</td><td colspan="4">FW[3:0]</td></tr>
<tr><td>18</td><td>12</td><td>RW</td><td>SFEN</td><td>LAGCEN</td><td colspan="2">-</td><td colspan="4">FWS[3:0]</td></tr>
<tr><td>20</td><td>14</td><td>RW</td><td colspan="4">-</td><td colspan="4">IOM[3:0]</td></tr>
<tr><td>21</td><td>15</td><td>RW</td><td colspan="2">-</td><td>SPIIPULL</td><td>SPIOPULL</td><td colspan="2">DRV[1:0]</td><td>OD012</td><td>OD3456</td></tr>
<tr><td>27</td><td>1B</td><td>RO</td><td colspan="8">ID[7:0]</td></tr>
<tr><td>28</td><td>1C</td><td>RO</td><td colspan="8">ID[15:8]</td></tr>
<tr><td>29</td><td>1D</td><td>RO</td><td colspan="8">ID[23:16]</td></tr>
<tr><td>30</td><td>1E</td><td>RO</td><td colspan="8">ID[31:24]</td></tr>
<tr><td>96</td><td rowspan="17">3</td><td>60</td><td>RO</td><td colspan="8">ANGLE [7:0]</td></tr>
<tr><td>97</td><td>61</td><td>RO</td><td colspan="8">ANGLE [15:8]</td></tr>
<tr><td>98</td><td>62</td><td>RO</td><td colspan="8">SPEED [7:0]</td></tr>
<tr><td>99</td><td>63</td><td>RO</td><td colspan="8">SPEED [15:8]</td></tr>
<tr><td>100</td><td>64</td><td>RW</td><td colspan="8">MULTITURN [7:0]</td></tr>
<tr><td>101</td><td>65</td><td>RW</td><td colspan="8">MULTITURN [15:8]</td></tr>
<tr><td>102</td><td>66</td><td>RO</td><td colspan="8">TEMP [7:0]</td></tr>
<tr><td>103</td><td>67</td><td>RO</td><td colspan="8">TEMP [15:8]</td></tr>
<tr><td>106</td><td>6A</td><td>RW</td><td>GPIODIR</td><td colspan="7">GPIO[6:0]</td></tr>
<tr><td>107</td><td>6B</td><td>RO</td><td colspan="8">READ COUNTER</td></tr>
<tr><td>108</td><td>6C</td><td>W1C/RO</td><td>MTP BUSY</td><td>RESERVED</td><td>INIT BUSY</td><td colspan="2">-</td><td>IFAIL</td><td colspan="2">RESERVED</td></tr>
<tr><td>111</td><td>6F</td><td>W1C/RO</td><td>IFB</td><td colspan="5">RESERVED</td><td>I2C</td><td>SPI</td></tr>
<tr><td>115</td><td>73</td><td>RO</td><td>-</td><td colspan="7">I2C SA</td></tr>
<tr><td>119</td><td>77</td><td>RW</td><td colspan="4">-</td><td colspan="3">ET_SHIFT[2:0]</td><td>ET</td></tr>
<tr><td>120</td><td>78</td><td>RO[7]
RW[6:0]</td><td>MTP DONE</td><td>-</td><td>STORE</td><td>RESTORE</td><td colspan="2">-</td><td colspan="2">RESERVED</td></tr>
<tr><td>126</td><td>7E</td><td>RO</td><td colspan="5">RESERVED</td><td>MTPLOCK</td><td>RESERVED</td><td>WRLOCK</td></tr>
<tr><td>127</td><td>7F</td><td>WO</td><td colspan="8">LOCK</td></tr>
</table>

Table 13: Factory Default Values

Add	Page	Hex	R/W	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0	0	00	RW	0	0	0	0	0	0	0	0
1		01	RW	0	0	0	0	0	0	0	0
2		02	RW	0	0	0	0	0	0	0	0
3		03	RW	1	1	1	1	1	1	1	1
4		04	RW	0	0	0	0	0	0	1	1
5		05	RW	0	0	0	0	0	0	0	0
6		06	RW	0	1	0	0	0	1	1	0
7		07	RW	0	0	0	0	0	0	0	0
11		0B	RW	0	0	0	0	0	0	0	0
12		0C	RW	0	0	0	0	0	0	0	0
15		0F	RW	0	x[1]	0	0	0	0	1	1
16		10	RW	0	0	0	0	0	0	0	0
17		11	RW	0	0	0	0	1	0	0	0
18		12	RW	1	1	0	0	1	0	0	0
20		14	RW	0	0	0	0	0	0	0	0
21		15	RW	0	0	0	0	0	0	0	0
106	3	6A	RW	0	0	0	0	0	0	0	0
119		77	RW	0	0	0	0	0	0	0	0

[1] x = 0 for SPI version (MA900GQE-S-0000), x = 1 for I2C version (MA900GQE-C-0000)

Table 14: Programming Parameters

Parameters	Symbol	Number of Bits	Description	See
Zero Setting	ZERO	16	Set the zero position	Table 18
Hysteresis	HYST	8	Hysteresis of the ABZ output	Table 23
Pulses Per Turn	PPT	11	Number of pulses per turn of the ABZ output	Table 21
Direction-A-Z Interface	DAZ	1	Direction / Increment / Index interface	Table 24
Index Length / Index Position	ILIP	4	Parametrization of the ABZ index pulse	Figure 44
Number of Pole Pairs	NPP	3	Number of pole pairs of the UVW output	Figure 48
Magnetic Field Low Threshold	MGLT	4	Sets the field difference low threshold	Table 35
Magnetic Field High Threshold	MGHT	4	Sets the field difference high threshold	
Push Button Interface Reset Time	PBI_TIME	3	Sets the reset time for Push Button Interface	Table 34
Push Button Interface Enable	PBI_EN	1	Enables the Push Button Interface	Table 32
PWM Mode	PWMM	1	Adds error detection to the PWM frame	Table 26
PWM Frequency	PWMF	1	Sets pulse width modulated output frequency	Table 28
SSI Parity	SSIP	1	Sets the SSI Parity Sign	Table 11
Reduced Wiring Mode	RWM	1	Enables the reduced wiring mode	Table 40
Serial Interface Switch	I2CEN	1	Enables I^2C interface instead of SPI	Table 4
Startup Write Lock	WLOCK	1	Set status of Write Lock at startup	Table 50
SPI Handshaking	SPIHS	1	Enables the SPI Handshaking check	Table 8
CRC Inversion	CRCINVEN	1	Enables CRC inversion when failure is detected	Table 7
CRC Enable	CRCEN	1	Enables CRC Check	Table 6
Rotation Direction	RD	1	Determines the sensor positive direction	Table 19
Filter Window	FW	4	Filter window for position output	Table 15
Speed Filter Enable	SFEN	1	Enables speed filter for lag compensation	Table 16
Lag Compensation Enable	LAGCEN	1	Enables the lag compensation	Table 17
Speed Filter Window	FWS	4	Filter window for speed output	Table 30
IO Matrix	IOM	4	Choice of the function of IO0-IO6 pins	Table 36
Pull Up/Down SPI Input	SPIIPULL	1	Determines the input circuits of /CS, SCLK and COPI pin	Table 44
Pull Down SPI output	SPIOPULL	1	Determines the output circuit of CIPO pin	Table 45
Drive Strength	DRV	2	Drive strength configuration for IO0-IO6 pins	Table 43
Open Drain Output IO0, IO1, IO2	OD012	1	Determines the output circuit of pins IO0, IO1, IO2: open drain or push-pull	Table 42
Open Drain Output IO3, IO4, IO5, IO6	OD3456	1	Determines the output circuit of pins IO3, IO4, IO5, IO6: open drain or push-pull	
ID	ID	32	Unique ID	
Angle	ANGLE	16	Angle output	

Parameters	Symbol	Number of Bits	Description	See
Speed	SPEED	16	Speed output	Table 29
Multiturn	MULTITURN	16	Multiturn output	
Temperature	TEMP	16	Temperature output	Table 31
GPIO Direction	GPIODIR	1	Configure direction of IO0-IO6 pins when configured as GPIO (input or output)	Table 37
GPIO Value	GPIO	7	Level of IO0-IO6 pins when configured as GPIO (input or output)	Table 38 Table 39
Read Counter	READ COUNTER	8	Read counter. Value incremented each time the register is read.	
MTP Busy	MTPBUSY	1	MTP Busy flag	Table 46
Init Busy	INITBUSY	1	Initialization Busy flag	
Interface Fault	IFAIL	1	Interface Fault flag	Table 47
Interface Busy	IFB	1	Register Read/Write while MTPBUSY = 1	
I2C Fault	I2C	1	I²C Fault flag	
SPI Fault	SPI	1	SPI Fault flag	
I²C Sensor Address	I2C SA	7	Sensor Address for I²C interface	Table 5
Electrical Test Shift	ET_SHIFT	3	Selects the Electrical Test angle	Table 20
Electrical Test Enable	ET	1	Enables the Electrical Test mode	
MTP Done	MTPDONE	1	MTP Store or Restore Completed	
MTP Store	STORE	1	Allows to perform a MTP Store Operation	
MTP Restore	RESTORE	1	Allows to perform a MTP Restore Operation	
MTP Write Lock	MTPLOCK	1	Enable write lock for MTP related registers	Table 49
Write Lock	WRLOCK	1	Enable write lock for all register	
Lock Sequence	LOCK	8	Sequence to toggle register locks	Table 48

REGISTER SETTINGS

DIGITAL FILTER

The signal provided by the sensor front-end is treated by a digital filter to improve the measurement resolution.

Filter Window Setting

The Filter Window (FW) parameter controls the sensor's angle output resolution (defined as the ±3σ noise interval), and cutoff frequency f_{CUTOFF}.

Table 15 provides the setting options and resulting performance. When FW = 0, the digital filter is disabled.

By default, the digital filter includes a strategy to cancel the lag when the magnet rotation speed is constant.

Table 15: Filter Setting

FW (3:0)	τ (µs)	Resolution[1] (bits)	Lag Cancellation	f_{CUTOFF} [2] (kHz)
0	0	9.8	No	100
1	2	8.0	Yes	200
2	4	8.7	Yes	125
3	8	9.4	Yes	65
4	16	10.2	Yes	29.5
5	32	10.7	Yes	13.5
6	64	11.3	Yes	6.44
7	128	11.7	Yes	3.15
8 (default)	256	12.4	Yes	1.56
9	512	12.9	Yes	0.776
10	1024	13.3	Yes	0.387
11	2048	13.8	Yes	0.193
12	4096	14.2	Yes	0.097
13	8192	14.4	Yes	0.048
14	16384	14.5	Yes	0.024
15	32768	14.5	Yes	0.012

Notes:

1) B_Z = 40 mT at sensing location
2) f_{CUTOFF} is defined as -3dB frequency on the response spectrum.

Transfer Function

To accurately model the system and analyze the stability of a control loop, both the Front-End and the Digital Filter transfer functions should be taken into account (see Figure 39).

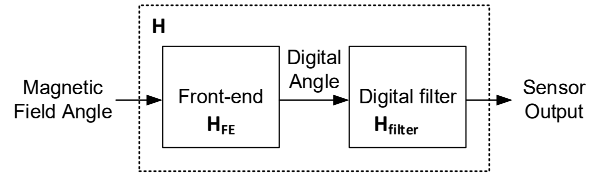


Figure 39: Transfer Function Diagram

The transfer function of the front-end signal is given as:

$$H_{FE}=\frac{1}{(1+\tau_{FE}\ s)^4} \quad (1)$$

The transfer function of the digital filter is:

$$H_{filter}=\frac{1+(2+\delta)\tau s}{(1+\tau s)^2} \quad (2)$$

Thus, the transfer function of the sensor is:

$$H=H_{FE}\ H_{filter} \quad (3)$$

With τ_{FE}= 1.6µs, $\delta=\frac{L}{\tau}$, and the time constants (τ) can be found in Table 15. If FW = 0, the filter is disabled and $H=H_{FE}$.

Figure 40 and Figure 41 show respectively the Bode magnitude and phase plot of H for FW values between 0 and 8.

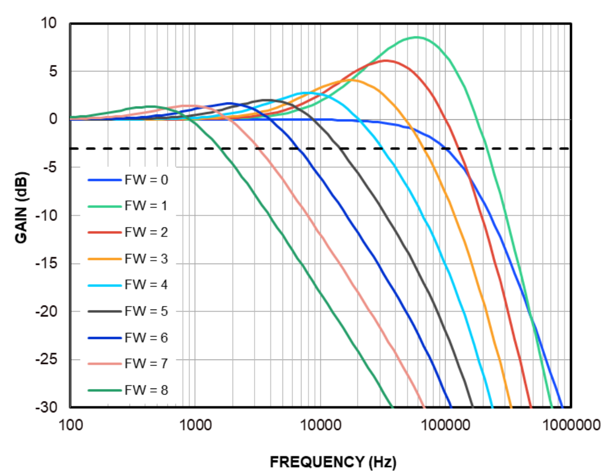


Figure 40: H Bode Gain Plot

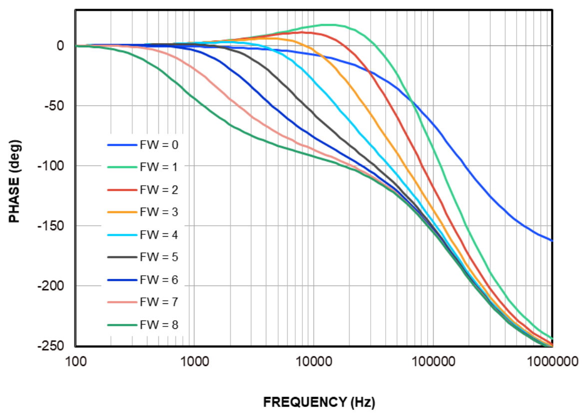


Figure 41: H Bode Phase Plot

Advanced Settings

As mentioned above, the MA900 digital filter implements by default a lag cancellation strategy.

This feature can be configured by changing SFEN and LAGCEN bits value in Register 18.

If SFEN = 1, the speed measurement is used by the digital filter to obtain a constant lag in the position measurement (Table 16).

Table 16: SFEN Setting

SFEN	Lag Compensation with Speed Measurement
0	Disabled
1 (default and recommended)	Enabled

If LAGCEN = 1, the lag compensation strategy is enhanced to obtain zero lag (Table 17).

Table 17: LAGCEN Setting

LAGCEN	Lag Cancellation
0	Disabled
1(default and recommended)	Enabled

It is recommended to keep SFEN = 1 and LAGCEN = 1. Please contact MPS if more information is required about the sensor performance outside of the recommended configuration.

ZERO SETTING

The zero position of the MA900 (a_0) can be programmed with 16 bits of resolution. The angle streamed out by the MA900 (a_{out}) is given by Equation (4):

$$a_{out} = a_{raw} - a_0 \quad (4)$$

Where a_{raw} is the raw angle provided by the MA900 digital filter.

The zero position is configured via the ZERO[15:0] parameter. The value of a_0 in degrees can be obtained as shown in Equation (5):

$$a_0 = 360\frac{\text{ZERO}[15:0]}{2^{16}} \quad (5)$$

Table 18 shows the zero-setting parameter.

Table 18: Zero-Setting Parameter

ZERO[15:0]	Zero pos. a_0 (deg)
0 (default)	0
1	0.005
2	0.011
…	…
65534	359.989
65535	359.995

Equation (6) shows a calculation example to set the zero position to 20°:

$$\text{ZERO}[15:0] = \frac{20°}{360°} \cdot 2^{16} = 3641 \quad (6)$$

ROTATION DIRECTION

By default, when looking at the top of the package, the angle increases when the magnet rotates clockwise (CW) (see Figure 42). The rotation direction can be changed by setting the RD field in Register 16 (see Table 19).

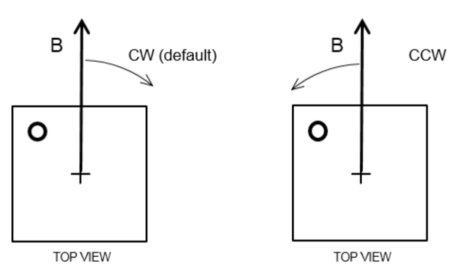


Figure 42: Positive Rotation Direction of the Magnetic Field. B indicates the field direction on the rotation axis

Table 19: Rotation Direction Parameter

RD	Positive Direction
0 (default)	Clockwise (CW)
1	Counterclockwise (CCW)

ELECTRICAL TEST MODE

The Electrical Test (ET) mode allows verifying the integrity of the sensor signal treatment. When enabled, by setting ET bit to 1, the signal from the Hall devices output is replaced by a predetermined reference signal. The system reverts to normal operation as soon as ET is set back to 0.

Via the ET_SHIFT bits, the reference signal can be configured to generate an angle which is a multiple of 45° (see Table 20).

Table 20: ET and ET_SHIFT Setting

ET	ET_SHIFT[2:0]	Angle Output
0	Any	Magnet angle
1	000	0°
1	001	45°
…	…	…
1	111	315°

ABZ INCREMENTAL ENCODER OUTPUT

The MA900 ABZ output emulates an incremental encoder (such as an optical encoder) providing logic pulses in quadrature (Figure 43). Compared to signal A, signal B is shifted by a quarter of the pulse period. The number of pulses per turn, N, sent on each signal can be configured, from 1 to 2048, via the parameter PPT, which consists of 11 bits split between Registers 3 and 4. Table 21 describes how to program PPT[10:0] to set the required ABZ output resolution.

Table 21: PPT Setting

PPT(10:0)	Pulses per Turn	Edges per Turn	
00000000000	1	4	MIN
00000000001	2	8	
00000000010	3	12	
00000000011	4	16	
…	…	…	…
11111111100	2045	8180	
11111111101	2046	8184	
11111111110	2047	8188	
11111111111	2048	8192	MAX

For example, to set 120 pulses per revolution (480 edges), set PPT to 120 – 1 = 119 (0b00001110111). Registers 3 and 4 must be set as shown in Table 22.

Table 22: Register 4 and Register 5

	B7	B6	B5	B4	B3	B2	B1	B0
R3	0	1	1	1	0	1	1	1
R4	-	-	-	-	-	0	0	0

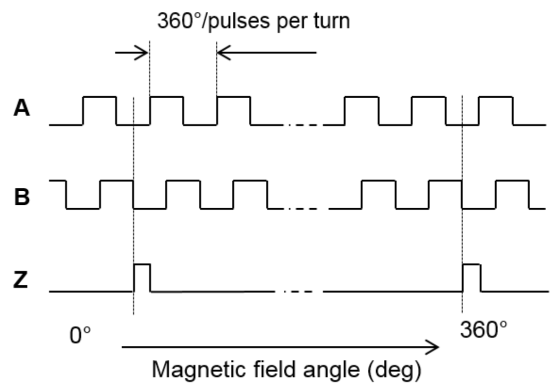


Figure 43: Timing of the ABZ Output

Signal Z (zero or index) raises only once per turn at the zero-angle position.

The position and length of the Z pulse is programmable via ILIP[3:0] parameter (see Figure 44).

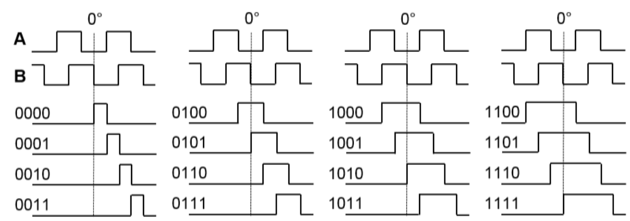


Figure 44: ILIP Parameter Effect on Index Shape

By default, ILIP is 0b0000. The Z rising edge is aligned with B falling edge and the index length is half the A or B pulse length.

ABZ Hysteresis

The hysteresis is set by the parameter HYST. If HYST[6:0] = 0, the hysteresis H (in degrees) is set to its minimum value:

$$H_{MIN} = \frac{360}{2^{16}} \quad (7)$$

If HYST[6:0] ≠ 0 and HYST[7] = 1, the hysteresis depends on FW field value.

If FW = 0:

$$H = \frac{360}{2^{15}} 2^{\frac{\mathrm{HYST}[6:0]-1}{12}} \qquad (8)$$

If FW ≠ 0:

$$H = \frac{360}{2^{15}} 2^{\frac{\mathrm{HYST}[6:0]+5-6\mathrm{FW}}{12}} \qquad (9)$$

In the latter case, H is always lower bounded by H_{MIN}.

If HYST[6:0] ≠ 0 and HYST[7] = 0, the hysteresis is:

$$H = \frac{360}{2^{15}} 2^{\frac{HYST[6:0]-1}{12}} \qquad (10)$$

Table 23 shows some H values for HYST[7] = 0.

Table 23: Hysteresis values for HYST[7] = 0

HYST[6:0]	H (deg)
0b0000000 (default)	0.0055
0b0000001	0.011
0b0000010	0.012
…	…
0b1111110	15.02
0b1111111	15.91

To avoid spurious transitions, it is recommended to set a hysteresis larger than 12 times the sensor's RMS noise (i.e., 12σ).

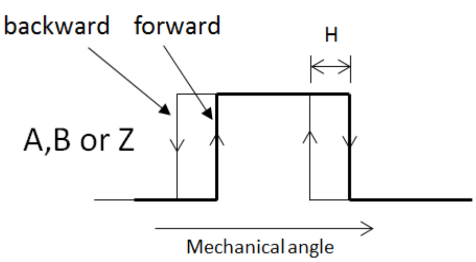


Figure 45: Hysteresis of the Incremental Output

ABZ Jitter

The ABZ signals are updated at 16MHz frequency, enabling accurate operation up to a very high speed (above 100krpm).

The jitter characterizes how far a particular ABZ edge can occur at an angular position different from the ideal position (see Figure 46).

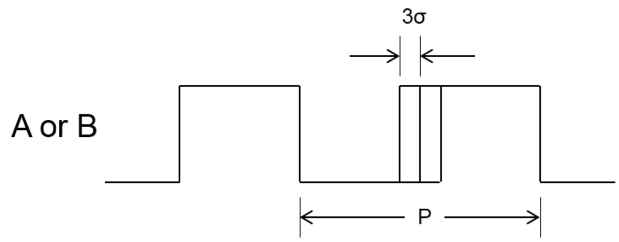


Figure 46: ABZ Jitter

The measurable jitter is composed of a systematic jitter (always the same deviation at a given angle) and a random jitter.

The random jitter reflects the sensor noise, so the edge distribution is the same as the SPI output noise. Like the sensor resolution, it is defined as the 3σ width of this distribution.

The random jitter is a function of the rotation speed. At higher speeds, the random jitter is smaller than the sensor noise. This is a consequence of the fact that, at high speed, the low frequency part of the noise spectrum does not contribute to the jitter.

DAZ Interface

For angular speed related applications, the ABZ interface can be converted to DAZ (D-direction / A-pulsing signal / Z-index) by programming DAZ bit to 1 (Table 24).

In this configuration, the D signal indicates CW (Logic 0) or CCW (Logic 1) rotation of the magnetic field, the A and Z signals remain unchanged compared to ABZ interface.

Table 24: DAZ Setting

DAZ	Incremental Encoder Output Format
0 (default)	ABZ
1	DAZ

UVW

The UVW output emulates the three Hall switches commonly used for block commutation of 3-phase electric motors. The three logic signals have a duty cycle of 50% and are shifted by 120° relative to each other (see Figure 47).

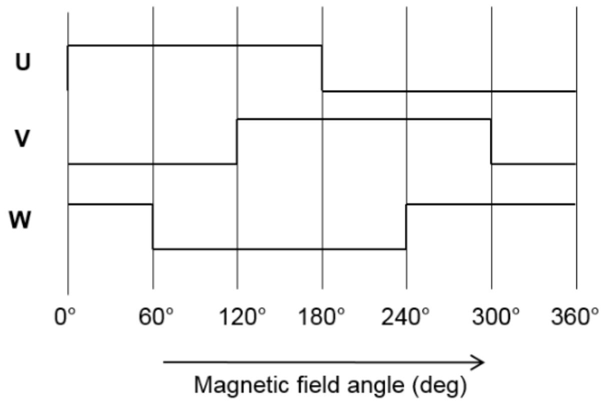


Figure 47: UVW output

The MA900 can emulate multiple pole pairs even if a target magnet with only 2 pole is used. This is achieved by dividing the digital angle into the required numbers of commutation steps per 360° revolution.

Parameter NPP[2:0] sets the number of pole pairs emulated, and the corresponding commutation step angle for the UVW signals. Table 25 describes the pole pairs configuration options.

Table 25: Number of UVW Pole Pairs

NPP [2:0]	Pole pairs	States per revolution	State width (deg)
000 (default)	1	6	60
001	2	12	30
010	3	18	20
011	4	24	15
100	5	30	12
101	6	36	10
110	7	42	8.6
111	8	48	7.5

Figure 48 shows the UVW signals when NPP = 2 is used.

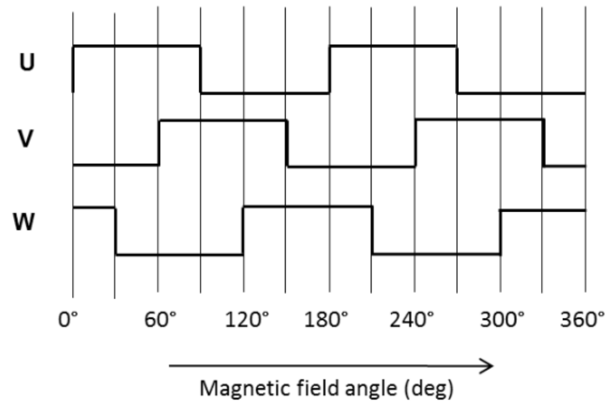


Figure 48: UVW signals for NPP = 2

UVW Hysteresis

A hysteresis larger than the output noise is introduced on the UVW output to avoid any spurious transitions (see Figure 49). The hysteresis value of UVW is also set by HYST (see Table 23) and is subject to the same recommendations.

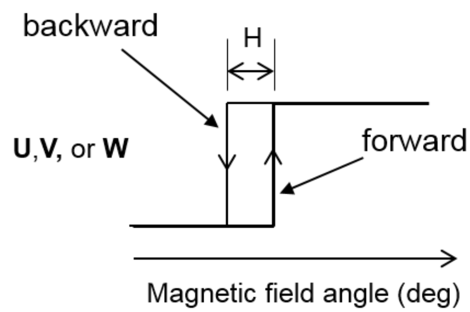


Figure 49: Hysteresis of the UVW Signal

PWM ABSOLUTE OUTPUT

The PWM output provides a logic signal whose duty cycle changes as a function of the measured angle.

The PWM period is divided into 4119 time units (counts). Each period is also composed by a series of 4 segments:

- Start, signal is held high.
- Error, signal is held high in normal operation (Figure 50). It is held low if PWMM = 1 and an error is detected (Figure 51).

- Data, in normal operation, signal is held high for a number of counts proportional to the measured angle, then it is driven low (Figure 50). The signal is kept low if PWMM = 1 and an error is detected (Figure 51).
- End, signal is held low.

It is possible to inform the controller about errors detected during the sensor operation, this option is enabled via PWMM bit (Table 26).

Table 26: PWM Mode selection

PWMM	Error transmission
0 (default)	Disabled
1	Enabled

Each segment has a fixed length in counts. The length of each segment is shown in Table 27.

Table 27: PWM segments durations in counts

	Start	Error	Data	End	Total
Counts	12	4	4095	8	4119

Two frequency options are available. The PWM frequency can be configured via PWMF bit (Table 28).

Table 28: PWM Frequency Selection

PWMF	PWM frequency
0 (default)	250 Hz
1	1 kHz

The recommended procedure to obtain the angle value is the following:

- Measure the PMW period t_{PWM} and the pulse width t_{HIGH} (Figure 50)
- Calculate the duration of a single count $t_{COUNT} = \frac{t_{PWM}}{4119}$
- Calculate the pulse width in counts $C_{HIGH} = \frac{t_{HIGH}}{t_{COUNT}}$
- The 12-bit angle is $Angle_{U12} = C_{HIGH} - 16$
- The angle in degrees is $Angle_{DEG} = \frac{Angle_{U12}}{2^{12}}$

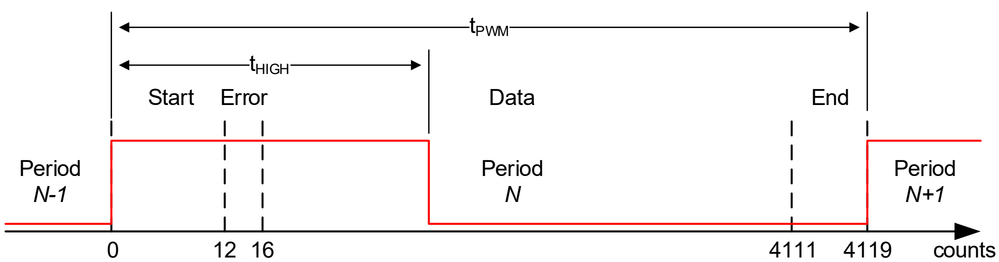


Figure 50: PWM output, normal operation

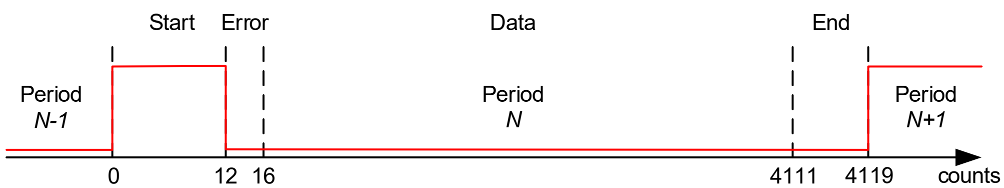


Figure 51: PWM output when PWMM = 1 and an error is detected

SPEED OUTPUT

Speed measurement is stored as a signed 16-bit decimal, in two's complement representation, in Registers 98 (bits 7 to 0) and 99 (bits 15 to 8) and it can be read either via SPI or I²C.

The speed measurement range is [-117187, 117183] rpm.

Equation (11) and Table 29 show the conversion from signed 16-bit decimal integer to rpm.

$$speed_{RPM} = speed_{S16}/3.576 \qquad (11)$$

Table 29: Speed Value Coding

Hexadecimal	Signed 16-bit	rpm
0x8000	-32768	-117187
0xFFFF	-1	- 3.57
0x0000	0	0
0x0001	1	+ 3.57
0x7FFF	32767	+117183

Speed output is derived from the position measurement and processed through a low-pass filter. FWS parameter in Register 18 controls resolution, time constant and cutoff frequency of the speed output, see Table 30.

Table 30: Speed Filter Setting

FWS (3:0)	Resolution (bits)	Time constant τ (μs)	f_{CUTOFF} (kHz)
0	Not recommended		
1	5.2	2	115
2	6.1	4	46.1
3	7.0	8	21.3
4	8.0	4	10.3
5	8.9	8	5.05
6	9.8	16	2.51
7	10.5	32	1.25
8 (default)	10.9	64	0.622
9	11.4	128	0.311
10	12.0	256	0.155
11	12.7	512	0.078
12	13.8	1024	0.039
13	14.7	2048	0.019
14	15.7	4096	0.009
15	16.0	8192	0.005

MULTITURN OUTPUT

The multiturn measurement is stored as a signed 16-bit decimal, in two's complement representation, in Registers 100 (bits 7 to 0) and 101 (bits 15 to 8) and it can be read either via SPI or I²C.

The multiturn measurement range is [-32768, 32767].

Whenever the angle of the MA900 crosses zero in positive direction (CW), the multiturn value increases by 1, and when it crosses zero position in negative direction (CCW), the multiturn value decreases by 1.

The current Multiturn value can also be changed by writing in Registers 100 and 101.

TEMPERATURE OUTPUT

Temperature measurement is stored as a signed 16-bit decimal, in two's complement representation, in Registers 102 (bits 7 to 0) and 103 (bits 15 to 8) and it can be read either via SPI or I²C.

The measured temperature range is from -40°C to 150°C.

Equation (12) and Table 31 show the conversion from signed 16-bit to °C.

$$temp_{°C} = temp_{S16} * 0.125 \qquad (12)$$

Table 31: Temperature value coding

Hexadecimal	Signed 16-bit	°C
0xFEC0	-320	-40
	…	
0xFFFF	-1	-0.125
0x0000	0	0
0x0001	1	+0.125
	…	
0x00C8	+200	+25
0x03E8	+1000	+125
0x04B0	+1200	+150

MAGNETIC FIELD DETECTION

MA900 provides two interfaces to monitor the field amplitude at the sensing location:

- MGH/MGL monitors if the field amplitude is inside a programmable range.
- PBI monitors the rate of change of the field amplitude, to detect events such as a button push/release.

Only one of the two interfaces can be active at a time. The user can select the active interface changing the PBI_EN bit value in Register 7 (see Table 32)

Table 32: PBI_EN Setting

PBI_EN	Active magnetic field detection interface
0 (default)	MGH/MGL
1	PBI

MGH/MGL

When PBI_EN = 0, the magnetic flags (MGH and MGL) are enabled. They indicate that the field amplitude detected by the sensor is outside a range defined by the lower (MGLT) and upper (MGHT) magnetic field thresholds.

The magnetic field thresholds are programmable (see Table 33). A 3mT hysteresis (MagHys) is introduced between the falling and rising edges of the MGL/MGH signals. Figure 52 shows the behavior of the MGL and MGH signals.

Table 33: Magnetic Thresholds

MGxT	Magnetic field threshold [mT]		
	MGH/MGL rising edge	MGH falling edge	MGL falling edge
0000	7	4	10
0001	14	11	17
0010	21	18	24
0011	28	25	31
0100	35	32	38
0101	42	39	45
0110	49	46	52
0111	56	53	59
1000	63	60	66
1001	70	67	73
1010	77	74	80
1011	84	81	87
1100	91	88	94
1101	98	95	101
1110	105	102	108
1111	112	109	115

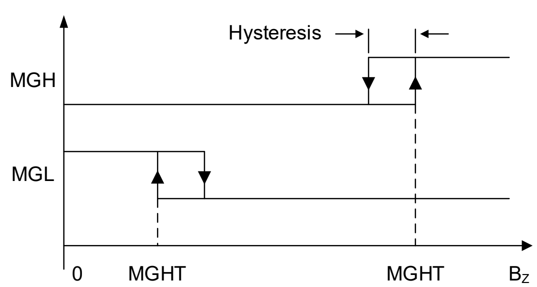


Figure 52: MGH and MGL Signals as a Function of the Field Amplitude

Every time either of the MGL or MGH outputs are high, the /FAULT pin is driven low.

PBI

When PBI_EN = 1, the Push Button Interface (PBI) signal is enabled. The PBI signal goes high every time a large enough increase of the field amplitude (e.g.: button push) is detected. On the other hand, the PBI signal goes low when a large enough decrease of the field amplitude (e.g.: button release) is detected.

To achieve this, the field amplitude signal goes through a high-pass filter, whose output is compared with configurable thresholds. Figure 53 shows the behavior of the PBI signal:

- At t_0 the button is pushed and the field amplitude increases
- At t_1 the high-pass filter output goes beyond the BTH threshold and the PBI signal goes HIGH
- At t_2 the button is released and the field amplitude decreases
- At t_3 the high-pass filter output goes below the BTL threshold and the PBI signal goes LOW

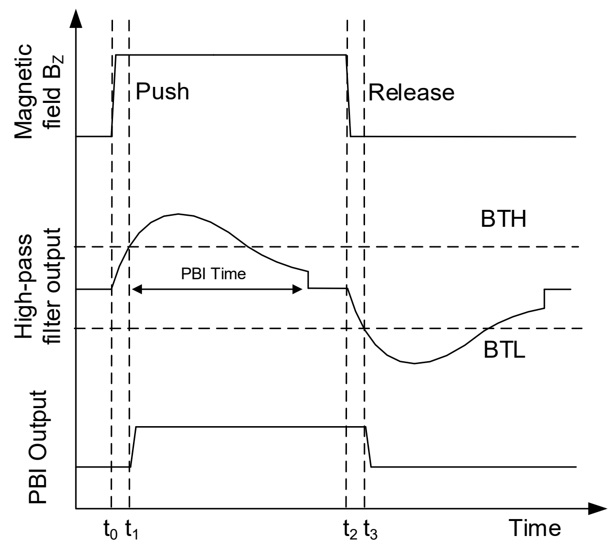


Figure 53: PBI Output

PBI_TIME parameter defines the high pass filter reset time. After a time PBI_TIME, after the PBI rising edge, the PBI analog signal is reset, allowing the detection to a new event, like the button release. Table 34 shows the configurable values of PBI_TIME.

Table 34: PBI_Time Setting

PBI_TIME	Delay between PBI rising edge and filter reset [ms]
000 (default)	2
001	4
010	8
011	16
100	32
101	65
110	131
111	262

Like the MGH/MGL case, the magnetic threshold can be configured by setting the MGLT/MGHT fields in Register 6 (see Table 35).

Table 35: PBI Thresholds

MGHT/MGLT	BTH [mT]	BTL [mT]
0000	2	-2
0001	2.5	-2.5
0010	3	-3
0011	4	-4
0100	5	-5
0101	6	-6
0110	7	-7
0111	8	-8
1000	9	-9
1001	10	-10
1010	11	-11
1011	12	-12
1100	13	-13
1101	14	-14
1110	15	-15
1111	16	-16

IO MATRIX

The IO output of the digital IO pins (IO1 to IO6) can be set by parameter IOM, as shown in Table 36.

Table 36: IO Matrix

IOM	IO0	IO1	IO2	IO3	IO4	IO5	IO6
0	Pull-Down						
1	A	B	Z	U	V	W	PWM
2	A	B	Z	/A	/B	/Z	PWM
3	A	B	Z	SSCK	SSD	PBI	PWM
4	SSCK	SSD	-	U	V	W	-
5	/U	/V	/W	U	V	W	-
6	A	B	Z	-	-	-	PBI
7	-	-	-	U	V	W	-
8	MGL	MGH	-	U	V	W	-
9	Reserved						
10	A	B	Z	MGL	MGH	-	-
11	-	-	PWM	-	-	-	PBI
12	MGL	MGH	-	SSCK	SSD	-	-
13	TRIG	-	-	-	-	-	-
14	-	-	-	-	-	-	-
15	GPIO0	GPIO1	GPIO2	GPIO3	GPIO4	GPIO5	GPIO6

TRIGGER

When IOM = 13, IO0 can be used as trigger input. When the trigger signal is high, the Angle, Speed, Multiturn and Temperature values are latched (not updated). This allows synchronized reading of multiple MA900 sensors.

GPIO MODE

GPIO Mode is selected when IOM=15. GPIODIR controls the direction of IO0-IO6 pads (Table 37).

Table 37: GPIODIR Setting

GPIODIR	GPIO Direction
0 (default)	Output
1	Input

When GPIODIR = 0, digital pins (IO0 to IO6) are configured as outputs. Writing in GPIO field defines the IO0-IO6 pins level. An example is shown in Table 38.

Table 38: GPIO Mode Output Example

GPIO[6:0]	IO Pin Level						
	0	1	2	3	4	5	6
0b0100110	0	1	1	0	0	1	0

When GPIODIR = 1, digital pins (IO0 to IO6) are configured as inputs. GPIO field provides the digital pins level. An example is shown in Table 39.

Table 39: GPIO Mode Input Example

IO Pin Level							GPIO[6:0]
0	1	2	3	4	5	6	
1	1	0	0	1	0	1	0b1010011

REDUCED WIRING MODE

The reduced wiring mode (RWM) allows to spare 3 connections on the PCB by reassigning the pins normally used by SPI/I^{2}C interfaces to a different functionality. RWM is enabled by setting RWM bit in Register 15 (see Table 40)

RWM register can only be written when MTPLOCK=0 (and WRLOCK=0), See Lock Mechanisms section for additional details.

Table 40: RWM Setting

RWM	Reduced Wiring Mode
0 (default)	Disabled
1	Enabled

To enable RWM the user shall:

1. Disable MTP Lock and Write Lock.
2. Set RWM = 1 in Register 15.
3. Perform a MTP Store (see MTP Operations).
4. Power cycle the MA900.

When RWM is enabled, pins 5, 6, 7 function is swapped with pins 8, 9, 10 (Table 41).

Significant care should be taken when planning to use RWM, as the operation is potentially irreversible. If pins 8, 9, 10 are left unconnected on the PCB, it is no longer possible to program the sensor once RWM is enabled.

Table 41: Pin Functionality in RWM Mode

RWM	Pin5	Pin6	Pin7	Pin8	Pin9	Pin10
0(default)	COPI	SCLK	CS	IO0	IO1	IO2
1	IO0	IO1	IO2	COPI	SCLK	CS

DIGITAL PIN I/O CIRCUIT

Digital pins (IO0 to IO6) are configured as push-pull by default when used as outputs (IOM = 15, GPIODIR = 0). These pins can be re-configured to an open drain behavior with the OD012 and OD3456 parameter (see Table 42, Figure 54). This allows to match systems with different signal levels (up to 5.5 V).

Table 42: Output pin configuration

OD012	IO0, IO1, IO2
0 (default)	Push-pull
1	Open Drain
OD3456	**IO3, IO4, IO5, IO6**
0 (default)	Push-pull
1	Open Drain

Figure 54 shows the internal circuits of the output pins IO1-IO6.

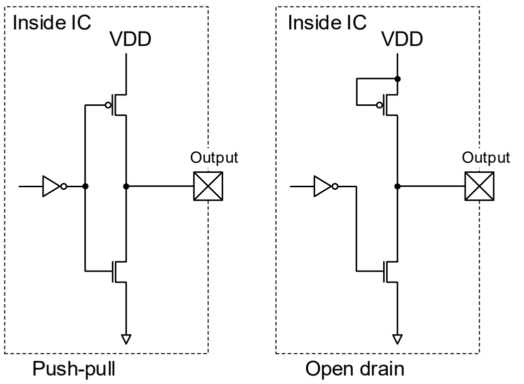


Figure 54: Internal Pin Structure for Push-Pull and Open Drain output options

The drive strength of digital outputs is configurable by using DRV setting (see Table 43)

Table 43: Drive strength

DRV	Drive strength I_O [mA]
00 (default)	4
01	8
10	12
11	16

SPI input pins are configured as high impedance by default. In this configuration, it is recommended to pull SCLK and COPI to GND; and CS to VDD to avoid any voltage buildup and inadvertent programming when the SPI interface is idle.

If required, the SPI inputs can be set to pull-up/pull-down configuration by the SPIIPULL parameter (see Table 44).

Table 44: SPI Input Pin Configuration

SPIIPULL	SPI input pins	
	CS	SCLK, COPI
0 (default)	High-Z	High-Z
1	Pull-up	Pull-down

Figure 55 shows the internal circuits of the SPI Input pins.

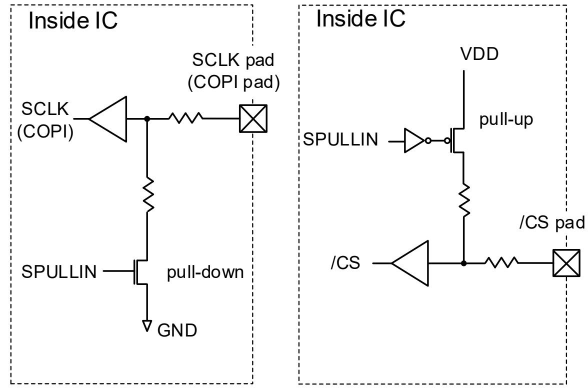


Figure 55: Internal Circuit for SPI Input Pins

The SPI output pin (CIPO) has tri-state logic. When SPI becomes idle, the CIPO is switched to high-impedance. This allows to connect multiple devices on the same SPI bus. If needed, CIPO can be pulled down when the SPI is idle by setting SPIOPULL = 1 (Table 45).

Table 45: SPI Output Pin Configuration

SPIOPULL	CIPO when SPI is idle
0 (default)	High-Z
1	Pull-down

Figure 56 shows the internal circuits of the SPI CS pin.

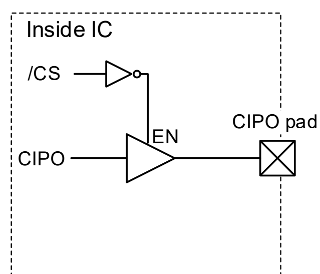


Figure 56: Internal Circuit for SPI CIPO Pin

BUSY FLAGS

Register 108 contains flags indicating specific conditions during the MA900 operation:

- MTPBUSY flag is set to 1 when either a MTP Store or MTP Restore operation is ongoing. The flag is automatically set to 0 when the MTP operation is completed.
- INITBUSY flag is set to 1 during the sensor initialization. It is automatically set to zero once the initialization is completed.

A summary of these flags is shown in Table 46.

FAULT DETECTION

As described in the SPI and I²C sections, MA900 is able to detect faults happening during the communication with the controller. There are three types of faults:

- IFB: a fault condition is generated if a SPI/I²C Write or Read Command is sent while an MTP Operation is ongoing (MTPBUSY = 1)
- I2C: a fault condition is generated if the CRC check is enabled (CRCEN = 1) and the controller does not send or sends a wrong CRC in an I²C Write Command.
- SPI: a fault condition is generated if the controller sends a wrong OpCode, CRC (if CRCEN = 1) or Handshaking (if SPIHS = 1).

Whenever a fault is detected, the following operations are performed:

- /FAULT pin is set to 0.
- IFAIL flag is set to 1 in Register 108.
- A fault-specific flag is set to 1 in Register 111.
- If SPI interface is enabled (I2CEN = 0) and SPI CRC inversion is enabled (CRCINVEN=1), the SPI CRC4 in SPI Communication is inverted.

Table 47 shows a summary of the MA900 fault related flags.

Table 46: Busy Flags

Flag	Register	Description
MTPBUSY	108	MTP operation ongoing
INITBUSY		Sensor Initialization ongoing

Table 47: MA900 Fault Related Flags

Flag	Register	Description
IFAIL	108	Interface fault detected
IFB	111	Read/Write access attempted while MTPBUSY = 1
I2C		I²C fault detected
SPI		SPI fault detected

All fault related flags in Registers 108 and 111 are sticky, namely they are not automatically cleared if the fault condition is removed. The flags must be cleared by writing a 1 in the related bit. The flag will not be cleared if the fault is persistent.

LOCK MECHANISMS

MA900 features two register write-lock options to prevent undesired modification to its registers:

- MTP Lock, register 120 is write-locked.
- Write Lock, all registers except 127 are write-locked.

A specific sequence of write commands (either SPI or I²C) is required to toggle the status of each lock. The sequence is composed by 3 ASCII characters to write consecutively in register 127 (see Table 48). When the sequence is received, the current lock access is toggled (from lock to unlock or vice versa). Any invalid character writes in Register 127 or any write at different address is cancelling the sequence.

Table 48: Lock ASCII Sequences

ASCII Sequence	Description
‘M’+’T’+’P’	Toggle MTP Lock
‘L’+’C’+’K’	Toggle Write Lock

The current status of the lock can be read on Register 126 (MTPLOCK and WRLOCK bits, see Table 49). MTP Lock is enabled by default, while WRLOCK is disabled by default.

Table 49: Lock Status Bits

MTPLOCK, WRLOCK	Lock Status
0	Disabled
1	Enabled

Additionally, the default/after-restore status of WRLOCK is configurable by using WLOCK field in Register 15 (see Table 50).

Table 50: WLOCK Setting

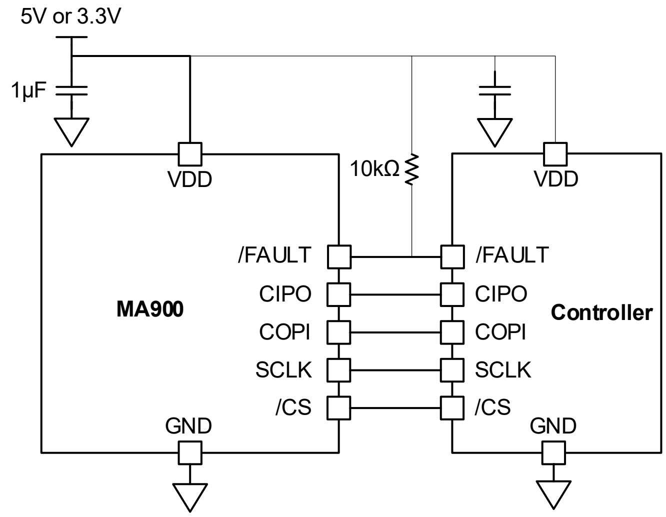
WLOCK	WRLOCK Default Status
0 (default)	Disabled
1	Enabled

UNIQUE ID

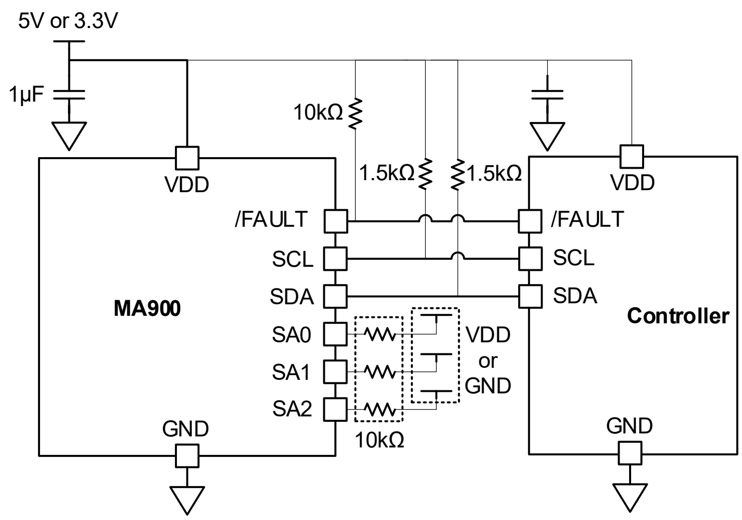
Each MA900 is provided with a unique, 32-bit ID code. The code can be obtained by reading Registers 27 to 30.

TYPICAL APPLICATION CIRCUIT

SPI

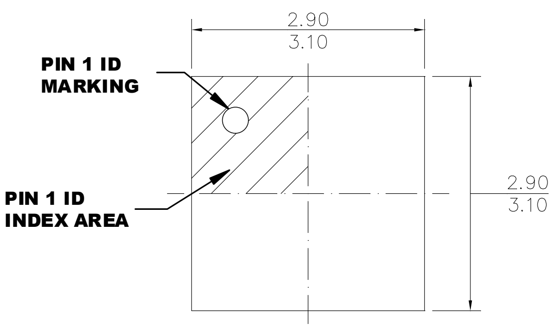


I²C

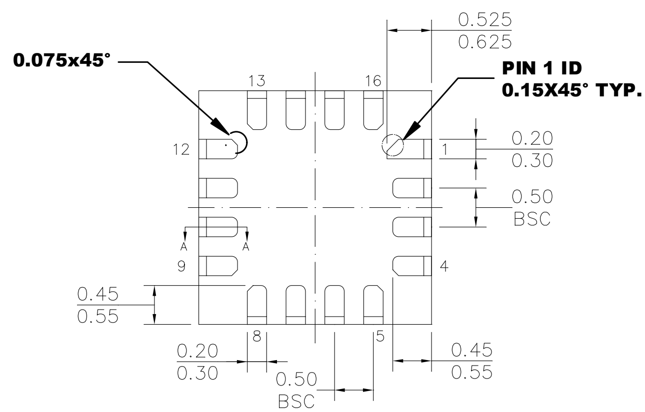


PACKAGE INFORMATION

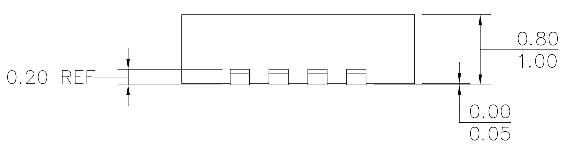
QFN-16 (3mmx3mm)



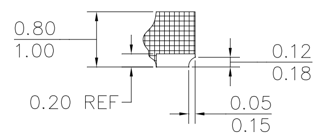
TOP VIEW



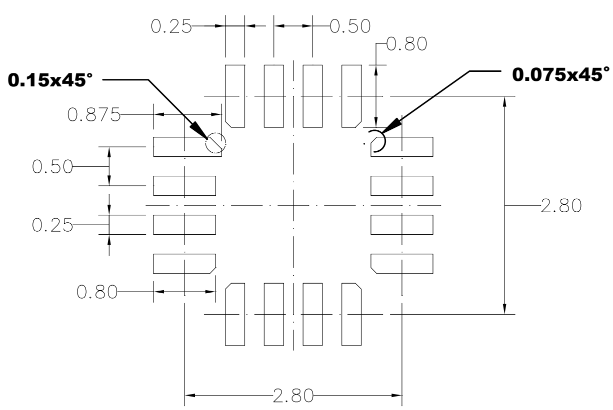
BOTTOM VIEW



SIDE VIEW



SECTION A-A

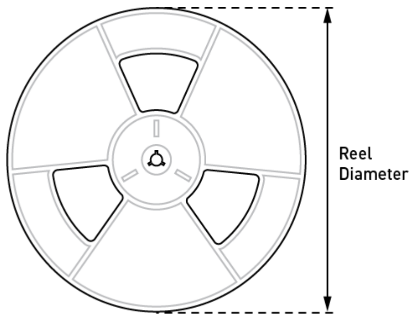


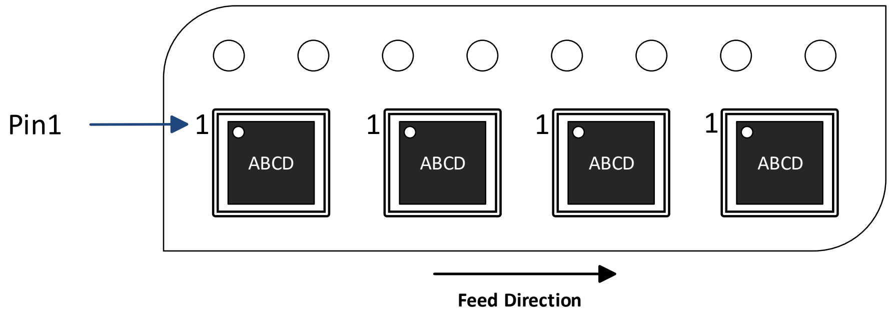
RECOMMENDED LAND PATTERN

NOTE:

1) THE LEAD SIDE IS WETTABLE.
2) ALL DIMENSIONS ARE IN MILLIMETERS.
3) LEAD COPLANARITY SHALL BE 0.08 MILLIMETERS MAX.
4) JEDEC REFERENCE IS MO-220.
5) DRAWING IS NOT TO SCALE.

CARRIER INFORMATION





Part Number	Package Description	Quantity/ Reel	Quantity/ Tube	Quantity/ Tray	Reel Diameter	Carrier Tape Width	Carrier Tape Pitch
MA900GQE-C-0000-Z	QFN-16 (3mm x 3mm)	5000	N/A	N/A	13 in.	12 mm	8 mm
MA900GQE-S-0000-Z							
MA900GQE-C-xxxx-Z							
MA900GQE-S-xxxx-Z							

APPENDIX A: CRC CALCULATION

CRC4 (SPI)

The CRC polynomial used for SPI is $x^4+x^3+x^2+1$, with an initial value of 10.

The C code below shows how to calculate the 16-bit value to send on COPI by appending the 4-bit CRC to the 12-bit data using a pre-computed CRC lookup table.

```
/* pre-computed CRC table for x^4+x^3+x^2+1 */

const uint8_t crc4_table[16] = {0, 13, 7, 10, 14, 3, 9, 4, 1, 12, 6, 11, 15, 2, 8, 5};

uint16_t append_crc4( uint16_t data ) {     // data is only 12-bit

 uint8_t crc4;                    // 4-bit CRC (0-15)

 // Compute CRC4

  crc4 = 10;                      // Initial Value for CRC4 (Seed)

  crc4 = (data>>8 & 0xF) ^ crc4_table[crc4];  // 4 MSB first

  crc4 = (data>>4 & 0xF) ^ crc4_table[crc4];  //

  crc4 = (data>>0 & 0xF) ^ crc4_table[crc4];  // 4 LSB

 // Concatenate 12-bit data with CRC4

 return (data<<4) | crc4;          // 16-bit word to send <12-bit data><4-bit crc>
```


CRC8 (I^2C)

The CRC polynomial used for I^2C is $x^8+x^2+x^1+x^0$ with an initial value of 0.

The C code below shows how to calculate the 8-bit CRC using a pre-computed CRC lookup table.

```
/* pre-computed CRC table for x^8+x^2+x^1+x^0 */

const uint8_t crc8_table [256] = {
       0x00,0x07,0x0e,0x09,0x1c,0x1b,0x12,0x15,0x38,0x3f,0x36,0x31,0x24,0x23,0x2a,0x2d,
        0x70,0x77,0x7e,0x79,0x6c,0x6b,0x62,0x65,0x48,0x4f,0x46,0x41,0x54,0x53,0x5a,0x5d,
        0xe0,0xe7,0xee,0xe9,0xfc,0xfb,0xf2,0xf5,0xd8,0xdf,0xd6,0xd1,0xc4,0xc3,0xca,0xcd,
        0x90,0x97,0x9e,0x99,0x8c,0x8b,0x82,0x85,0xa8,0xaf,0xa6,0xa1,0xb4,0xb3,0xba,0xbd,
        0xc7,0xc0,0xc9,0xce,0xdb,0xdc,0xd5,0xd2,0xff,0xf8,0xf1,0xf6,0xe3,0xe4,0xed,0xea,
        0xb7,0xb0,0xb9,0xbe,0xab,0xac,0xa5,0xa2,0x8f,0x88,0x81,0x86,0x93,0x94,0x9d,0x9a,
        0x27,0x20,0x29,0x2e,0x3b,0x3c,0x35,0x32,0x1f,0x18,0x11,0x16,0x03,0x04,0x0d,0x0a,
        0x57,0x50,0x59,0x5e,0x4b,0x4c,0x45,0x42,0x6f,0x68,0x61,0x66,0x73,0x74,0x7d,0x7a,
        0x89,0x8e,0x87,0x80,0x95,0x92,0x9b,0x9c,0xb1,0xb6,0xbf,0xb8,0xad,0xaa,0xa3,0xa4,
        0xf9,0xfe,0xf7,0xf0,0xe5,0xe2,0xeb,0xec,0xc1,0xc6,0xcf,0xc8,0xdd,0xda,0xd3,0xd4,
        0x69,0x6e,0x67,0x60,0x75,0x72,0x7b,0x7c,0x51,0x56,0x5f,0x58,0x4d,0x4a,0x43,0x44,
        0x19,0x1e,0x17,0x10,0x05,0x02,0x0b,0x0c,0x21,0x26,0x2f,0x28,0x3d,0x3a,0x33,0x34,
        0x4e,0x49,0x40,0x47,0x52,0x55,0x5c,0x5b,0x76,0x71,0x78,0x7f,0x6a,0x6d,0x64,0x63,
        0x3e,0x39,0x30,0x37,0x22,0x25,0x2c,0x2b,0x06,0x01,0x08,0x0f,0x1a,0x1d,0x14,0x13,
        0xae,0xa9,0xa0,0xa7,0xb2,0xb5,0xbc,0xbb,0x96,0x91,0x98,0x9f,0x8a,0x8d,0x84,0x83,
        0xde,0xd9,0xd0,0xd7,0xc2,0xc5,0xcc,0xcb,0xe6,0xe1,0xe8,0xef,0xfa,0xfd,0xf4,0xf3
      };

uint8_t crc8(uint8_t crc8_curr,  uint8_t data ) { // inputs are current CRC value and data
  crc8 =crc8_table[crc8_curr ^ data];  // table index is XOR between current CRC and new data
 return crc8;
```


The following code shows how the controller must calculate the CRC for an I²C Write Single operation:

```
uint8_t sensor_address = 0x62;

uint8_t register_address = 0x00;

uint8_t write_value = 0x05;       // value to write

uint8_t crc = 0;

crc = crc8(crc, sensor_address << 1);

crc = crc8(crc, register_address);

crc = crc8(crc, write_value);
```

If an I²C Write Burst is performed, controller must send the CRC after each value. Assuming the values are stored in an array, for each step i, the CRC is calculated as follows:

```
uint8_t crc = 0;

crc = crc8(crc, sensor_address << 1);

crc = crc8(crc, register_address + i);

crc = crc8(crc, write_value[i]);
```

In case of a I²C Read Single, the CRC sent by the sensor is calculated as follows:

```
uint8_t sensor_address = 0x62;

uint8_t register_address = 0x00;

uint8_t read_value;        // value received

uint8_t crc = 0;

crc = crc8(crc, sensor_address << 1);

crc = crc8(crc, register_address);

crc = crc8(crc, (sensor_address << 1) + 1);

crc = crc8(crc, read_value);
```

If a I²C Read Burst is performed, the same principle of I²C Write Burst applies.
Finally, for an I²C Quick Read, for each step i, the CRC is calculated as follows (please note that the first register address in this case in 96):

```
uint8_t crc = 0;

crc = crc8(crc, (sensor_address << 1) + 1);

crc = crc8(crc, register_address + i);

crc = crc8(crc, write_value[i]);
```